A MULTI-MODAL FRAMEWORK FOR QUANTIFYING NARRATOR RELIABILITY IN ʿILM AR-RIJAL USING EVALUATIVE STANCE AND GRAPH NEURAL NETWORKS

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2026

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MATRIC NUMBER : BI23110344

TITLE **:** A MULTI-MODAL FRAMEWORK FOR QUANTIFYING NARRATOR RELIABILITY IN ʿILM AR-RIJAL USING EVALUATIVE STANCE AND GRAPH NEURAL NETWORKS

DEGREE **:** BACHELOR OF COMPUTER SCIENCE WITH HONOURS

(DATA SCIENCE)

VIVA'S DATE : DD SEPTEMBER 2027

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DECLARATION

I hereby declare that the material in this thesis is my own except for quotations, equations, summaries and references, which have been duly acknowledged.



30 APRIL 2026

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ACKNOWLEDGEMENT

"خَيْرُ النَّاسِ أَنْفَعُهُمْ لِلنَّاس"

"*The best of people are those who are most beneficial to others.*"

– Prophet Muhammad ﷺ

(At-Tabarani in Al-Muʿjam al-Awsat Hadith: 5787, graded *Ṣaḥīḥ* by Al-Albani)

All praise and gratitude belong to Allah for giving me the strength, patience, and clarity to complete this proposal. This work is a small attempt to serve the tradition of ʿIlm ar-Rijal, and I pray that it benefits others in some way.

I would like to express my deepest thanks to my supervisor, Dr. Khalifa Chekima, for his guidance, patience, and honest feedback throughout this journey. He gave me the freedom to explore an ambitious idea while quietly making sure I didn't go off the rails.

I am also grateful to the researchers and developers behind the OpenITI corpus, Shamela Digital Library, and CAMeL Tools, as their work made this project possible in the first place. Likewise, I thank the authors of the AR-Sanad 280K dataset and the Ifta' Sunnah platform for making high-quality Arabic Hadith data accessible to the research community.

To my parents and family: thank you for your endless prayers, for believing in me and for never doubting in my study years. This degree is as much yours as it is mine.

Finally, to my friends who listened to me ramble about Fiqh, madhabs and *dalils*, thank you for the many "spinning" sessions and for the generous number of ideas we all made.

May Allah accept this small effort of mine, forgive my shortcomings, and make it beneficial to those who study His religion.

Alan Syahmi Bin Sanih @ Sani

8th May 2026

ABSTRACT

The authentication of Prophetic traditions (Hadith) through biographical narrator evaluation (*ʿIlm ar-Rijāl* and *al-Jarḥ wa t-Taʿdīl*) is a foundational discipline of Islamic scholarship, yet manual cross-referencing across classical compendia struggles to scale with digital repositories. Furthermore, existing computational approaches focus predominantly on entity disambiguation or suffer from target data leakage and untested graph homophily assumptions. This thesis presents a zero-leakage and multi-modal computational framework for quantifying narrator reliability across three experimental stages. First, data from the Hadith-Viz biographical encyclopaedia and the SanadSet transmission corpus for the Six Canonical Hadith Collections (*al-Kutub as-Sittah*) are processed through a five-stage data hygiene funnel and a Dual Temporal Firewall. This firewall purges post-classical consensus verdicts (7th–9th centuries AH) to isolate primary critical statements from early authorities (*al-Mutaqaddimūn*, 2nd–4th centuries AH), mapping targets to Ibn Ḥajar al-ʿAsqalānī's canonical hierarchy via parameter-free Min-Max normalisation ($0.0,1.0$) and three ordinal tiers (*Thiqah*, *Ṣadūq*, and *Below Ṣadūq*). Second, across the text benchmark cohort ($N = 819$, evaluated over 25 paired stratified cross-validation folds), regularised lexical TF-IDF logistic regression achieves a Macro-F1 of 0.6852 ± 0.0514 (80.59% accuracy, $QWK = 0.6697$), significantly outperforming frozen AraBERT v2 contextual embeddings ($Macro - F1 = 0.5635 \pm 0.0373$, $p = 3.49 \times 10^{-11}$) due to the formulaic lexical specificity of classical criticism. Third, a 13-arm controlled ablation study on the multimodal graph cohort ($N = 335$ nodes, 177 undirected transmission edges) demonstrates that Tabular Early Fusion MLP achieves the highest multimodal Macro-F1 (0.4810 ± 0.0684), whereas Graph Convolutional Network (GCN) message passing degrades performance ($Macro - F1 = 0.4311 \pm 0.0750$, $p = 0.0215$) due to structural network heterophily ($h = 0.4107$) across teacher-student links. Finally, downstream validation across canonical chains using non-compensatory weakest-link aggregation ($S_{\text{chain}} = \min_i \hat{y}_i$) acts as the evaluation protocol. This establishes an explainable, epistemologically grounded decision-support tool for Hadith scholarship.

Keywords: Hadith authentication, *ʿIlm ar-Rijāl*, *al-Jarḥ wa t-Taʿdīl*, evaluative stance analysis, AraBERT, Graph Neural Networks, network heterophily, multi-modal framework

ABSTRAK

RANGKA KERJA MULTIMOD UNTUK MENGUANTIFIKASI KEBOLEHPERCAYAAN PERAWI DALAM ILMU RIJAL MENGGUNAKAN ANALISIS PENDIRIAN EVALUATIF DAN RANGKAIAN NEURAL GRAF

Pengesahan hadis melalui ilmu rijal dan *al-jarḥ wa al-ta'dīl* berdepan kekangan rujukan silang manual berskala besar, manakala model pengkomputeran sedia ada sering terdedah kepada kebocoran data sasaran serta andaian homofili graf yang tidak diuji. Kajian ini mencadangkan satu rangka kerja multimod bebas kebocoran bagi mengkuantifikasi kebolehpercayaan perawi berlandaskan korpus biografi Hadith-Viz dan transmisi SanadSet bagi *al-Kutub as-Sittah*. Saluran data dibersihkan menerusi Dinding Api Temporal Dwi-Lapisan bagi mengasingkan kenyataan pengkritik awal (*al-Mutaqaddimūn*, abad ke-2–4 H) daripada konsensus pascaklasik, lalu dipetakan kepada tiga tahap kebolehpercayaan ordinal (*Thiqah*, *Ṣadūq*, dan *Bawah Ṣadūq*) berpandukan taksonomi Ibn Ḥajar al-ʿAsqalānī. Merentasi penanda aras teks ($N = 819$), model TF-IDF leksikal mencapai Macro-F1 0.6852 ± 0.0514, mengatasi pembenaman AraBERT v2 (0.5635, $p < 0.001$) secara signifikan ekoran kekhususan leksikal istilah teknikal klasik. Dalam eksperimen ablasi multimod 13-lengan ($N = 335$), model *Early Fusion MLP* mencatat Macro-F1 tertinggi (0.4810 ± 0.0684), manakala penyampaian mesej GCN menurunkan prestasi (0.4311, $p = 0.0215$) berpunca daripada heterofili struktur rangkaian ($h = 0.4107$) pada pautan guru-murid. Akhir sekali, pengesahan hiliran ke atas isnad kanonik menggunakan pengagregatan pautan terlemah ($S_{\text{chain}} = \min_i \hat{y}_i$) bertindak sebagai protocol penilaian. Ini membuktikan keupayaan sistem sebagai alat sokongan keputusan yang berwibawa dan berteraskan epistemologi hadis.

Kata kunci: Pengesahan hadis, *'Ilm ar-Rijāl*, *al-Jarḥ wa al-Ta'dīl*, analisis pendirian evaluatif, AraBERT, Rangkaian Neural Graf, heterofili rangkaian

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CHAPTER 1

INTRODUCTION

1.1 Overview

In Islam, rulings on a matter are divided into two. The first few, around 5% of total rulings, are called *qat'i* 'definitive' as they are found to be explicitly mentioned in the Quran. The rest are grouped as *dhanni* 'speculative' rulings, which are primarily derived from *ijtihad*, a trained Islamic jurist's interpretation (Kamali, 2014). To ground their interpretation as truthfully as possible, they rely on a secondary corpus after the Quran called hadith, literally meaning 'report, narration' (Ahmed & Anas, 2025). However, due to the hadith's nature being a reported or narrated saying, in other words comparable to modern day telephone game, each narrator in the chain (*sanad*) of the said hadith has to be assessed and graded to determine the hadith's grade, from *sahih* 'strong' to *da'if* 'weak (Kamali, 2014). Once this process is done, its text content (*matn*) which describes the words or action of Prophet Muhammad will be critically assessed. These two core components, *sanad* 'chain' and *matn* 'content' makes what a hadith is. Due to this speculative nature of hadith, Muslims worldwide would be at odds on the status of permissibility or impermissibility of most secondary matters, i.e. new issues that are not mentioned explicitly in the Quran (Kamali, 2014).

This poses two information challenges for hadith verification. The first, a hadith entirely depends on the people who passed it down, i.e. the narrators in a *chain* (*sanad*). Secondly, the same people must be considered as honest and accurate by their contemporaries. These challenges naturally lie at the intersection of network analysis (graph machine learning) and text processing (NLP) within the domain of computer science. To preserve any notes, comments or remarks, whether praise or criticism, by contemporaries of a narrator, later scholars compiled them into massive biographical compendiums, and this domain is called *'Ilm ar-Rijal*, literally 'the Study of Men'. Any praise or criticism of a narrator will be weighted by the same later scholars, then gave their verdict on a narrator's trustworthiness, from which

this domain is known by its name *'Ilm al-Jarh wa t-Ta'dil*, literally 'the Study of Criticism and Praise' (Ahmed & Anas, 2025; Ismail, 2024).

The digitisation of classical texts through global repositories in recent years has made the Hadith corpus accessible on an unprecedented scale. Yet paradoxically, this very accessibility has created a new challenge: the capacity of modern scholarship to rigorously authenticate and verify Hadith at the speed of digital dissemination has not kept pace with the volume of circulation (Mosa, 2025). The manual cross-referencing labour of biographical entries across hundreds of classical volumes remains an expert-dependent process that is resistant to scale, presenting a compounding challenge for the field (Mosa, 2025).

In hopes of bridging this gap, this project develops a multi-modal and data-driven framework that automates and streamlines the process of narrator reliability evaluation while remaining grounded in the classical principles of *Jarḥ wa Taʿdīl*. The framework integrates three complementary computational techniques:

i. Evaluative Stance Analysis: Extracting and classifying qualitative appraisals of integrity and precision from classical texts using lexical n-grams and pretrained Arabic transformer models (Antoun et al., 2020);
ii. Graph Machine Learning and Heterophily Analysis: Modelling narrator transmission networks to see if mentor-disciple graphs show structural heterophily or if relational topology offers a discriminative signal (Lim et al., 2021; Zhu et al., 2020); and
iii. Continuous and Ordinal Reliability Quantification: Combining classical three-tier ordinal classifications with aggregated historical assessments to create a standardised dependability index ($0.0 - 1.0$) (Ghriss, 2024).

The goal of this project is to use modern data science approaches to create a reproducible and epistemologically grounded system that directly supports the classical discipline of *ʿIlm ar-Rijāl*.

1.2 Problem Background

Historically, classical Study of Criticism and Praise began in between roughly 150 and 350 AH, by foundational critical hadith experts such as Yaḥyā ibn Maʿīn, Aḥmad ibn Ḥanbal, al-Bukhārī, and Abū Ḥātim ar-Rāzī in which they put individual transmitters against two non-negotiable criteria, Integrity (*ʿadālah*) and Precision (*ḍabṭ*) (Ahmed & Anas, 2025; Ismail, 2024). Integrity is defined by moral uprightness and honesty, ensuring they are free from fraud or any bias. Conversely, precision is defined by memory retention and manuscript preservation,

guaranteeing that texts were maintained without confusion, omissions, or accidental additions. For the classical experts, piety alone was insufficient, as an upright and pious narrator would still be classified as weak if their notebooks were found to be disordered or their memory has faded away with age (Ahmed & Anas, 2025; Ismail, 2024).

As centuries goes by, early historical records have fragmented and scattered across personal notebooks, local city libraries, and student lecture notes. Additionally, new sects emerged with new doctrinal biases, leading to scholars concluding differing narrator verdicts. All this results in manual cross-referencing slower and demands heavy scholarly training. Nonetheless, a new generation of hadith experts also emerged that managed to compile the fragmented records into comprehensive biographical works complete with their own 'narrator verdict harmonisation' from earlier verdicts. A notable example is *Taqrīb al-Tahdhīb*, authored by the 15th century hadith expert Ibn Ḥajar al-ʿAsqalānī (al-ʿAsqalānī, 1995). In his work, the earlier records are condensed and harmonised into verdicts with a 12-tier ordinal hierarchy based on his own critical analysis. This is considered to be one of the early efforts to standardise the categories of verdicts made in the Study of Criticism and Praise.

This historical corpus presents a rare intersection of rich biographical text paired with modern data science through an extensive relational mentor-student transmission graph method. Recent studies have also demonstrated the feasibility of applying machine learning to Hadith data (A'thoina et al., 2025), primarily for named entity recognition (NER) and narrator disambiguation across *sanad* graphs (Ahmad et al., 2025; Mosa, 2025; Muther et al., 2023). The biographical works do not behave like standard sentiment data, as concise and technical labels are used, and simply adding one qualification e.g. *ṣadūq yahim* 'truthful, he makes mistakes' will flip the reliability tier entirely.

Furthermore, recent graph-based workflows face two unresolved methodological issues: target data leakage and untested homophily assumptions. Prior models often extracted text features without purging 8th- and 9th-century consensus verdicts from input records, risking data leakage where models memorise target labels rather than learning evaluative stance. Additionally, standard Graph Neural Networks assume homophily, meaning connected nodes share similar labels (McPherson et al., 2001). In hadith transmission networks, however, celebrated master scholars routinely taught students of widely varying mnemonic competence, creating heterophilous networks where naive graph message passing risks degrading classification accuracy (Lim et al., 2021; Zhu et al., 2020).

1.3 Problem Statements

Three primary and significant challenges remain in the computational study of Hadith despite promising advancements in Arabic NLP:

i. Extraction Without Quantification (Identification vs. Judgment): While current models have improved in Named Entity Recognition (NER) for extracting narrator names and lineages, there is a critical lack of computational frameworks capable of quantifying narrator reliability. Identifying a narrator is insufficient, as the framework must be able to "judge" their standing based on historical evidence (Ahmad et al., 2025).

ii. Graph Models Stop at Structural Extraction: Recent work by Mosa (2025) and Oumer (2025) has shown that transformer models are capable of extracting narration chains from Hadith text as graphs. Nonetheless, those approaches stop at the graph, as they did not assign a reliability score to any narrator or use any actual *Jarḥ wa Taʿdīl* praise or criticism terms to judge narrator trustworthiness. This research goes beyond graph construction by adding an evaluative reliability index and GNN-based structural analysis.

iii. Target Data Leakage and Untested Graph Homophily: Prior computational workflows frequently extracted text features from biographical compendia without filtering out target consensus grades, introducing target data leakage. Moreover, existing systems assume transmission graphs exhibit homophily without empirical verification; in reality, teacher-student links often connect reliable masters to weak students, meaning standard GNN message passing can cause over-smoothing and harm predictive performance (Lim et al., 2021; Zhu et al., 2020).

1.4 Project Objectives

The main objectives of this project are as follows:

i. To develop an NLP classification model that maps qualitative scholarly judgments (*Jarḥ wa Taʿdīl*) to a standardised, quantitative reliability index ($0.0 - 1.0$) and a 3-tier ordinal classification under strict zero-leakage conditions.

ii. To construct an audited Knowledge Graph of narrator transmission relationships and execute a controlled multi-arm ablation study to investigate whether structural graph features improve prediction or suffer from transmission network heterophily.

iii. To evaluate the association between narrator reliability predictions aggregated over complete transmission chains and independently reviewed hadith grades using weakest-link scoring.

1.5 Project Scope

To ensure that the research remains focused, this project operates within a clearly defined set of boundaries, a technically feasible methodology, and is evaluable within the constraints of an undergraduate final year project.

1.5.1 Data Sources

The framework is trained and tested on structured biographical narratives, transmission networks, and digitised classical Arabic texts. Biographical records, narrator metadata, and teacher-student associations are sourced from the curated Hadith-Viz biographical encyclopaedia, which indexes classical biographical materials from the Shamela digital library, specifically the Taqrīb al-Tahdhīb of Ibn Ḥajar al-ʿAsqalānī and the Tahdhīb al-Kamāl of al-Mizzī. These classical compendia contain the most comprehensive collections of narrator appraisals and establish the ground-truth benchmark for Jarḥ wa Taʿdīl consensus labels.

1.5.2 Scale and Dataset

The experimental scope targets three primary cohorts drawn from canonical Hadith collections (*al-Kutub as-Sittah*):

i. A primary biographical text cohort of several hundred canonical non-Companion narrators for stance modelling and zero-leakage text benchmarking;

ii. A multimodal sub-cohort possessing verified biographical quotations, network transmission edges, and canonical grades for the controlled ablation study; and

iii. A candidate set of historical transmission chains for downstream evaluation, subject to narrator identity review and independent grade verification.

1.5.3 Technical Scope

The project is organised around three interconnected tasks:

i. Evaluative Stance Extraction: Processing historical critic quotations using lexical n-gram baselines and pretrained Arabic transformer models (such as AraBERT);
ii. Entity and Relational Modelling: Modelling teacher-student relationships via a Knowledge Graph and evaluating Graph Neural Networks (such as GCNs) against non-relational baselines; and
iii. Reliability Scoring & Planned Chain Evaluation: Mapping multi-critic evaluations to a continuous index $S \in [0.0, 1.0]$and 3-tier ordinal grades. Additionally, narrator scores will be aggregated across complete reviewed chains using a weakest-link operator and compared with independently reviewed grades.

1.5.4 Application Domain Limitations

a) Exclusion of the Prophet's Companions (*Ṣaḥābah*): In orthodox Sunni Islamic methodology, all Companions are recognized as universally upright (*kulluhum ʿudūl*) and are exempt from critical disparagement (*jarḥ*). Including Companions would artificially inflate majority-class metrics and distort the machine learning objective.

b) Exclusion of Content (*Matn*) Criticism: This research is strictly concerned with the structural and biographical evaluation of transmission chains (Sanad criticism). The analysis of Hadith textual content (*Matn* criticism) requires comparative jurisprudential reasoning across Sunni legal schools and falls outside the scope of this project.

c) Decision Support (No Theological Autonomy): Any automated reliability score produced by the framework is intended as a decision-support tool for academic research, not as a definitive Islamic ruling or fatwa. The epistemological authority of qualified human scholars (*muḥaddithūn*) is expressly preserved.

d) Computational and Hardware Boundaries: All experiments are executed locally in batch on consumer hardware. Generative large language models and prompt-based classification are deliberately excluded to ensure deterministic, reproducible embeddings and strict adherence to target leakage controls.

1.5.5 Out of Scope

The following are explicitly excluded from this project:

a) *Matn* (textual body) analysis or content-based Hadith grading;
b) Coverage of Shīʿa isnād evaluation methodologies;
c) Real-time or web-based deployment of the scoring pipeline; and
d) Languages other than Arabic (e.g., Urdu or Persian Hadith commentaries).

1.6 Organisation of Report

The remainder of this report is structured as follows:

- **Chapter 2 (Literature Review):** Reviews the classical science of *ʿIlm ar-Rijāl*, surveys existing computational hadith systems and Arabic language models, and examines graph representation learning and the challenge of graph heterophily.

- **Chapter 3 (Project Methodology):** Details the multi-critic corpus construction, the zero-leakage temporal filtering protocol, graph construction, and model architectures.

- **Chapter 4 (System and Experiment Design):** Outlines the experimental matrix, baseline specifications, ablation configurations, and evaluation metrics.

- **Chapter 5 (Implementation):** Documents the software environment, preprocessing modules, training pipelines, and reproducibility parameters.

- **Chapter 6 (Results and Analysis):** Presents completed empirical findings for PO1 and PO2 and reports the status of the PO3 evaluation.

- **Chapter 7 (Conclusion):** Concludes the findings and significance of this project.

CHAPTER 2

LITERATURE REVIEW

2.1 The Development of ʿIlm ar-Rijāl and Biographical Criticism

Hadith primarily depends on its *isnād* 'chain'. In Islamic learning, the process of authentication of a report involves checking each of the narrators who reported it; this led to the complementary sciences of ʿIlm ar-Rijāl (literally the Study of the Men [of the isnāds]) and ʿIlm al-Jarḥ wa t-Taʿdīl, the Study of Criticising and Appraising the names of Hadith narrators.

2.1.1 Historical Developments of Evaluative Dictionaries

Narrator criticism, initially an oral tradition in 2nd century AH, originated and was engaged in by critics like Shuʿbah ibn al-Ḥajjāj (d. 160 AH) and Sufyān ath-Thawrī (d. 161 AH), who frequently subjected transmitters to a verbal test of credibility and memory. After three phases spanning over the next three centuries, this oral criticism evolved into written biographical dictionaries:

i. **The formative phase (2nd–3rd century AH):** Some early reviewers listed personal reviews and notes on transmission on their personal books. These notes were collected by ʿAlī ibn al-Madīnī (d. 234 AH), Aḥmad ibn Ḥanbal (d. 241 AH), and Yaḥyā ibn Maʿīn (d. 233 AH).

ii. **Systematisation (3rd–4th century AH):** Muḥammad ibn Ismāʿīl al-Bukhārī (d. 256 AH) wrote *al-Tārīkh al-Kabīr* that arranged the transmitters alphabetically. Soon afterward Ibn Abī Ḥātim al-Rāzī (d. 327 AH) wrote *al-Jarḥ wa t-Taʿdīl* that established a ranking of transmitters for disapproval and approval.

iii. **The encyclopaedic synthesis phase (7th–9th century AH):** Al-Mizzī (1980) created a "biographical-historical synthesis" in *Tahdhīb al-Kamāl fī Asmāʾ ar-Rijāl* that recovered all biographical material from the six books of *al-Kutub as-Sittah*. Adh-Dhahabī (1992) condensed these biographical narratives in *Mīzān al-Iʿtidāl* and *al-*

Kāshif. Ibn Ḥajar al-ʿAsqalānī composed *Tahdhīb al-Tahdhīb* (al-ʿAsqalānī, 1984) and its twelve-table brief, *Taqrīb al-Tahdhīb* (al-ʿAsqalānī, 1995).

Modern hadiths historians such as Shaykh ʿAbdullāh al-Judaīʿ (2003) argue that it is important to distinguish that there is a significant historiographical distinction between the early critics (*al-Mutaqaddimūn*, 2nd–4th centuries AH) and late compilers (*al-Mutaʾakhkhirūn*, 7th–9th centuries AH). The early critics were contemporary to the transmitters who were observed to be present at audition sittings (*samāʿ*), heard oral variants as they happened, and offered primary empirical evidence. The late compilers such as Ibn Ḥajar had no such privileged eyewitness account; they only read earlier conflicting testimonies to establish a jurisprudential synthesis of a consensus verdict.

This temporal division shapes the machine learning pipeline for this project. Criticism comments from the beginning are used only as input text features, whereas consensus grades from the 9th century Ibn Ḥajar are used as the supervised target labels. Our pipeline commits to this temporal partition and abolishes target data leakage.

2.1.2 The Two Criteria of ʿAdālah and Ḍabṭ

In classical hadith methodology a narrator's trustworthiness is not ascribed as a binary label. It is instead rooted in two separate and independent criteria:

i. **Integrity (*ʿAdālah*):** The narrator is able to differentiate fact from fiction, is of good moral character, adheres to the faith, and is truthful. Serious deviations include open bad character (*fisq*), sectarian prejudice (*bidʿah*), or fabrication (*kidhb*). The transmission is then identified as abandoned (*matrūk*) or fabricationist (*kadhdhāb*).

ii. **Precision (*aḍ-Ḍabṭ*):** The narrator must prove an accurate memory and meticulous preservation of the manuscript. Researchers differentiate *aḍ-ḍabṭ* into retention of memory (*ḍabṭ aṣ-ṣadr*) and preservation of the manuscript (*ḍabṭ al-kitāb*). Inability of *aḍ-ḍabṭ* could be neglect (*ghaflah*), extreme distortion from authentic messengers (*fuḥsh al-ghalaṭ*), or old age impairment (*ikhtilāṭ*).

Classical critics felt that by itself piety was not enough to ensure dependability. If an upright and pious narrator misplaced his notebooks or confused names due to old age, critics would lower his transmission rank.

2.1.3 Hierarchical Categorisation Schemes

In *Taqrīb al-Tahdhīb*, Ibn Ḥajar al-ʿAsqalānī established a classic taxonomy of twelve types of grade (al-ʿAsqalānī, 1995). These range, in the context of modelling, into three canonical ordinal classes:

i. **Tier 2 (*Thiqah* / Trustworthy):** The full set of grades of Ibn Ḥajar that designates his highest level of accreditation, including *Thiqah Thiqah* (eminently trustworthy) and *Thiqah* (trustworthy and precise). However, *Ṣaḥābah* are removed from machine learning part of this project, as their very high ethical and moral integrity is already presumed traditionally.

ii. **Tier 1 (*Ṣadūq* / Truthful):** Narrators who have clear absence of moral defect and partial memory shortcomings, e.g., *ṣadūq yahim* 'truthful, he makes mistakes'. Their narrations form the backbone of good (*ḥasan*, below authentic) hadiths.

iii. **Tier 0 (*Below Ṣadūq* / Unreliable or Qualified):** Transmitters who, due to unverified identity (*majhūl*), weak memory (*ḍaʿīf*), severe unreliability (*matrūk*), or proven fabrication (*kadhdhāb*), are not clearly trustworthy.

This classification makes evaluative stance computationally available for the machine to classify and makes it distinct from the more general concept of sentiment.

2.2 Classical Evaluative Stance Analysis vs. Generic Sentiment Analysis

Biographical assessment dictionaries (*kutub ar-rijāl*) are not amenable to standard sentiment analysis lexicons. *Al-Jarḥ wa t-Taʿdīl* discourse is not emotional affective text, but a specialised forensic language register for communicating evaluative stance.

2.2.1 The Concept of Evaluative Stance in Biographical Criticism

Evaluative stance represents the linguistic indication of an author's epistemic perspective, evidential certainty, or normative stance through linguistic cues (Biber & Finegan, 1988). In biographical criticism, evaluative cues demonstrate the narrator's classification of accuracy, transmission, and retention. It uses technical, evidence-based appraisal as opposed to a general emotional feeling.

2.2.2 Structural Failure of Generic Sentiment Dictionaries

Applying generic Arabic sentiment dictionaries to biographical texts results in many incorrect classifications due to three domain-specific language reasons:

i. Domain-specific polarity inversion: In common usage, *fāḍil* (virtuous), *zāhid* (ascetic), or *rajul ṣāliḥ* (pious man) have positive connotations. In *rijāl* work, they describe an individual's piety, but do not confirm retention mnemonically (*aḍ-ḍabṭ*). Early critics often noted, "He was a pious man, but careless in his recording of hadith," so while a typical sentiment model registers this as strongly positive, the classical criteria would resolve the transmitter as weak.

ii. Lexically ordered compound qualifiers: Critics habitually position positive and negative words together to produce fragile composite qualifiers. Sentences like *ṣadūq illā annahu yukhṭi'* ('truthful, except that he makes mistakes') or *lā ba'sa bihi fī ghayr ḥadīth az-Zuhrī* ('acceptable, except when transmitting from az-Zuhrī') consist of qualifying clauses that must be syntactically parsed instead of relying on simple keyword bag-of-words techniques.

iii. Granular technical gradation: Classical descriptors sit at specific semantic points along a scale. For example, *thiqah* is known for well-established reliability, *ṣadūq* is reliable to a tolerable degree with a middle-range memory, *maqbūl* (acceptable) needs corroboration, and *ḍa'īf* denotes active unreliability.

2.3 Computational Methods in Hadith and Isnād Analysis

Computational study of hadith books has made significant progress in the last decade. Early studies were concerned mainly with text digitisation and creating searchable relational datasets (Saloot et al., 2016). Once a structured corpus was available, researchers began deploying sequence models for Named Entity Recognition (NER) by extracting narrator tokens directly from the raw text. More recent studies reached into graph theory to interpret network topologies for transmission chains.

The limits of data availability really set the computational constraints. When FYP1-based investigations were carried out, machine-learning available biographical collections simply did not exist; the preliminary work were based almost entirely on crude and unannotated digital library dumps of Shamela, from which only through ingenious regular expressions and heuristics could viable biographical reports be derived from multi-volume texts. It was incredibly brittle, filled with old print and typesetting, containing nested editorial notes, and marked by inconsistent scholar boundary conventions. However recently, Hadith-Viz has assemble canonical transmitters into verified entity records, with itemised judgments of historical criticisms and educational references, had been released. The latter development

shifts the computational challenge from inconsistent scrapes and print bugs to proper structure inference and evaluation modelling. But two common blind spots can be found in current literature.

Firstly, existing pipelines end at identity resolution. They determine the who, or if two experts might have ever met, but do not test whether the transmitter even makes a credible narrator. Link prediction models such as Narrator2Vec (Mghari et al., 2024) can find transmission co-occurrence, but they give an equal structural consideration to any edge. An encounter between two masters is represented the same as an audition with an untrustworthy writer. Reliability was altogether overlooked.

Second, end-to-end hadith grading models circumvented narrator-level forensics. When classifiers simply reasoned about authenticity of whole transmission chains as strings of text (Ahmad et al., 2025), there was 100% confounding: one fabricated report in a five-transmitter chain leaves the entire transmission chain provided as input to appear superficial, not fabricated, so the individual transmitter is erased. How can a flattened sequence model reason about which transmitter is to blame here, leaving us with inscrutable black boxes that defy classical verification logic?

Table 2.1 summarises, compares and contrasts the key computational approaches in this field and compares them against this project, taking into account network size, evaluation time and method flaws.

Table 2.1: Critical Comparative Benchmark of Prior Computational Literature

Study	Primary Approach and Network Architecture	Target Task	Dataset and Scale	Primary Metric	Evaluated Performance	Methodological Deficiencies Relative to This Work
Mahmoud et al. (2022)	Deep neural sequence matching on synthetic	Narrator entity disambiguation in	AR-Sanad 280K	Accuracy,	91.2% Accuracy	Evaluated only on artificial chains, which

	sanad graphs	chain sequences	artificial chains	Macro-F1		were extracted using rule-based templates; biographical text and reliability of narrators were not evaluated.
Mahmoud et al. (2024)	Graph query embedding + AraBERT re-ranking over *NarratorsKG*	Narrator entity detection in multi-hop graphs over *NarratorsKG*	AR-Sanad 280K-v2 dataset	Top-1 Accuracy	94.6% Top-1 Accuracy	Focused on detecting the entity rather than quantifying reliability; evaluations are done over synthetic graphs rather than canonical historical collections.
Mghari et al. (2022, 2024)	Continuous skip-gram and CBOW narrator sequence embeddings	Corpus extraction (SanadSet) and missing link prediction	SanadSet (650,986 raw isnād chains)	Cosine similarity, Link AUC	Captures transmission co-occurrence	Functions only on syntactic chain co-occurrence; ignores classical

	(*Narrator2Vec*)					biographical evaluation texts entirely, with no means to output reliability scores.
Ahmad et al. (2025)	Sequence tagging (adapting HMM isnād structures from Najeeb, 2022) + supervised classifiers on flattened isnāds	Chain continuity flagging and direct binary hadith grading (*Ṣaḥīḥ* vs *Ḍa ʿīf*)	Canonical *Kutub as-Sittah* sample	Accuracy, Precision, Recall	75% to 82% Accuracy	Bypasses individual narrator modelling; depends on a dataset-wide correlation of grades for each isnād, creating a black box that cannot explain why an isnād is false.
Ghriss (2024)	Bipartite graph co-update algorithm modelling multi-path trust propagation	Information authentication and collective trust propagation	Curated mathematical toy sub-graphs	Ranking correlation	Theoretical convergence proved	Provides an algorithmic proof-of-concept on toy graphs; does not parse real classical Arabic

						biographical texts.
Mosa (2025)	Hybrid knowledge graph and transformer architecture for entity linking	Historical narrator entity disambiguation in classical manuscripts	Curated historical isnād records	Precision, Recall, F1	93.8% F1-score	Resolves onomastic naming ambiguity; does not extract evaluative stance or model qualitative reliability tiers.
Proposed Framework (This Project)	Multi-modal architecture that fuses lexical n-grams, contextual transformer embeddings, and transmission graph representations	Narrator reliability measurement and trust at chain-level	Canonical *Kutub as-Sittah* biographies and transmission graphs	Macro-F1, Quadratic Weighted Cohen's κ	Formally assessed with cross-validation benchmarks	Surpasses previous constraints through rigorous zero-leakage temporal filtration, stance extraction at the narrator-level, and empirical examination of transmission network homophily.

2.4 Pretrained Arabic Language Models

Arabic NLP has pretty much moved away from word-level static embedding backbones toward deep contextual transformers. Static embeddings (Word2Vec, FastText, for example) give each lexeme an invariant vector, which works very poorly for Semitic languages. For Arabic, because grammatical markers, conjunctions, and prepositions are attached to lexical stems as clitics, static embeddings suffer from a major limitation: individual lemmata are just about impossible to use for classical text. Static vectors just do not work.

2.4.1 Arabic NLP Transformer-Based Models

Contextual transformer architectures (Vaswani et al., 2017) disambiguate polysemy through multi-head self-attention on the context of the entire sequence. In Arabic NLP the effectiveness of the model depends on whether or not tokenisation preserves the morphological complexity of the language:

a) **AraBERT (Antoun et al., 2020):** Based on BERT-Base (12 transformer layers, 768 hidden layer size, 12 attention heads, 110M parameters). To deal with agglutination in Arabic, AraBERT used the Farasa segmenter for pre-processing before subword tokenisation. This segmenter separates clitics, conjunctions (*wa-*), prepositions (*bi-*, *li-*), and definite article pre-consonant vowels to increase vocabulary coverage in both old and formal Arabic texts.

b) **CAMeLBERT (Inoue et al., 2021):** Evaluated the effect of linguistic distance by learning distinct models for Classical Arabic (CA), Modern Standard Arabic (MSA), and dialects. Results show that transformer-based language models pretrained on a historical classical corpus drastically outperform models pretrained on modern journalistic news content in terms of perplexity on medieval Islamic texts.

2.4.2 Linguistic Obstacles in Classical Biographical Text

Evaluating modern Arabic models on 2nd–4th century evaluative treatises uncovers serious domain mismatch during pretraining. Most pretrained models were trained on news feeds, Wikipedia articles, or social media text. When used directly on classical biographical dictionaries, three language phenomena harm the quality of representations:

a) Syntactic nesting and honorific parentheticals: Classical biographical articles often add long salutes (*tarāḥum*, for instance *raḥimahu Allāh*, and *ta ʿẓīm*) between the person

mentioned in the biographical note and the critic's evaluation. Honorific parentheticals include the critical modifiers but place the referent several dozen tokens away.

b) Lexical semantic change (Diachronic semantic drift): Words change meaning considerably from the time of the classical writers to the present day. For example, the adjective *ṣāliḥ* refers generally to piety in present terms, but was employed by the early hadith critics to mean an acceptable transmission.

c) Subword splitting of composite appraisals: General-purpose subword tokenisers break apart frequent lexical combinations into separate units (for example *laysa bi-qawiyy* 'not strong' or *lā ba'sa bihi* 'no harm in him'). Rare expressions are also split into separate subword units, separating the evaluative sentiment expressed by idiomatic evaluation compounds.

2.5 Graph Neural Networks and the Graph Heterophily Problem

A class of deep learning methods that operate on non-Euclidean topologies by updating features of individual nodes through an iterative process of neighbourhood aggregation.

2.5.1 Foundations of Graph Representation Learning

Consider a transmission network $G = (V, E)$, where V represents the set of N narrators (nodes) and Erepresents confirmed master-student transmission relations (edges). In a standard GCN, the node representation matrix $H^{(l)}$ at layer l updates via localised first-order spectral convolution (Kipf & Welling, 2017),:

$$H^{(l+1)} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\tilde{A}\widetilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$

Equation 2.1

Here, $\tilde{A} = A + I_N$ represents the adjacency matrix augmented with self-loops, $\widetilde{D}_{ii} = \sum_j \tilde{A}_{ij}$ is its degree matrix, $W^{(l)}$ is a trainable transformation matrix, and $\sigma(\cdot)$ is the element-wise activation function. For the two-layer GCN in this work, the update is:

$$Z = \widetilde{D}^{-\frac{1}{2}}\tilde{A}\widetilde{D}^{-\frac{1}{2}}\mathrm{ReLU}\left(\widetilde{D}^{-\frac{1}{2}}\tilde{A}\widetilde{D}^{-\frac{1}{2}}XW_0\right)W_1$$

Equation 2.2

where $W_0 \in \mathbb{R}^{8\times64}$and $W_1 \in \mathbb{R}^{64\times3}$ for the PO2 graph-only model. Graph Attention Networks use a different approach to compute dynamic attention coefficients α_{ij} across connected neighbours (Veličković et al., 2018).

2.5.2 The Assumption of Homophily and Its Limitation

Graph neural network (GNN) architectures commonly encode a sociological assumption: graph homophily (McPherson et al., 2001). Nodes connected to each other are more likely to belong to the same class:

$$h = \frac{|\{(u,v) \in E : y_u = y_v\}|}{|E|}$$

Equation 2.3

When edge homophily is high ($h \to 1$), recursive message passing is a handy low-pass filter of a graph signal. Here, local averaging smooths intra-class differences and increases the accuracy of classification (Kipf & Welling, 2017).

2.5.3 The Phenomenon of Graph Heterophily in Transmission Networks

When a network exhibits strong heterophily ($h \ll 0.5$ for binary tasks, or $h \leq 1/C$ in multi-class settings with C classes), connected nodes are mostly from different classes. In heterophilous cases, conventional GNN message passing and aggregation would further decrease classification accuracy (Zhu et al., 2020; Lim et al., 2021). Neighbour aggregation would over-smooth the high-diversity class distribution, with node embeddings collapsing into a single centroid. Performance is destroyed.

Structural heterophily is an unavoidable historical fact of hadith transmission networks. It originates directly in the classical pedagogical process:

a) **Renowned master scholars (*shuyūkh*):** Masters such as Sufyān ath-Thawrī or Imām Mālik held public evaluations to scores of their aspirants in hundreds of sittings. These circles comprised extremely careful scholars amongst slack ones.

b) **Scholarly travels (*riḥlah fī ṭalab al-ḥadīth*):** Seekers of hadith were willing to travel thousands of miles within the Islamic realm to collect and preserve rare chains from minor local elders.

Edges therefore record pedagogical encounters of the past, not shared mnemonic accuracy. The combination of educational contact and shared reliability in transmission causes

both truthful and untruthful signals to become indistinct, thus discouraging narrator classification.

2.6 Summary and Identification of Research Gaps

Three basic research gaps on the intersection of computational linguistics and hadith verification are identified from the existing literature:

2.6.1 The Semantic-Structural Separation

Existing works split their efforts. They examine transmission graphs devoid of the conditioning text (Mghari et al., 2024; Ghriss, 2024), or do entity linking without measuring the reliability for biographic records (Mahmoud et al., 2024; Mosa, 2025). There is no framework combining contextual Arabic language embeddings with the topology of the transmission graph to yield a reliable prediction of narrator grades.

2.6.2 Methodological Vulnerability and Target Data Leakage

Earlier predictive models could rely on the consensus verdicts of the 9th century as features, which resulted in artificially high scores of classification. Moreover, several evaluations used raw Accuracy as a performance indicator in such an imbalanced setting that it hid the prediction error rates of the minority class.

2.6.3 Unexamined Graph Homophily Assumptions

Previous work on GNNs for transmission networks has assumed that graph convolution would improve prediction, without checking whether *isnād* networks are in fact homophilic. No prior work ran controlled statistical ablations to find out if message passing would enhance or detract from narrators' reliability quantification. The field does not have an empirical benchmark.

CHAPTER 3

PROJECT METHODOLOGY

3.1 Research Design

The research study uses an experimental research design within the DSRM framework. The main aim of the current research study is to design and empirically validate a multi-scale leakage free computational pipeline for measuring narrator reliability in ʿIlm ar-Rijāl by using the combination of contextual text modelling and graph representation learning. The full framework architecture and multi-scale cohort pipeline are shown in the following Figure 3.3 (see in Section 3.5). In order to investigate the key questions defined in Chapter 1, the methodology has been divided into the three main experimental stages:

i. Stage 1 (Data ingestion, source auditing and preprocessing): Canonical Hadith transmission chains and biographical evaluation corpora are ingested, audited for corruption, and subjected to a five-stage data hygiene funnel with a Dual Temporal Firewall for zero target leakage.
ii. Stage 2 (Transmission Graph Construction and Feature Engineering): An empirical transmission network is constructed from the canonical *isnād* chains; as a result of this graph construction, teacher-student relations are turned into an undirected graph and a corresponding topological and demographic feature tensor with 8 feature channels.
iii. Stage 3 (Reliability Normalisation, Multimodal Modelling and Planned Downstream Evaluation): Quantitative classical assessments are translated into continuous reliability measures through a parameter-free Min-Max transformation. Text modelling and multimodal graph fusion are assessed for PO1 and PO2 experiments; and finalising with PO3 chain-level assessment.

3.2 Stage 1: Data ingestion, source auditing & preprocessing

3.2.1 Sources of data

The experimental pipeline is made of two main classical Hadith data sources:

i. Data from Mghari et al. (2024), with raw isnād chains from classical Hadith collections (650,727 chains). This study limits transmission corpus to the 30,712 chains corresponding to the six canonical collections (al-Kutub as-Sittah): Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, Sunan Abī Dāwūd, Jāmiʿ at-Tirmidhī, Sunan an-Nasāʾī, and Sunan Ibn Mājah.
ii. Biographies, historical information and judgments were extracted from the Hadith-Viz biographical encyclopaedia (Dataset/hadithviz/shamelarijal2.json), which includes information about 3,047 classical scholars, including biographical information, teacher-student relationships, date of death, location, and generation (ṭabaqāt), *al-Jarḥ wa t-Taʿdīl* (list of judgments per item), etc.

3.2.2 Forensic Audit of Scraped Primary Texts

Previously, exploratory workflows tried to scrape biographical records out of raw digitised volumes of Tahdhīb at-Tahdhīb, using regular expression scraping (output/shamela/). A forensic audit was conducted on the scraped output which found that poor data corruption had occurred.

Of the extracted biographical blocks, 51.9% contained other-than-person narrative segments, poetic references, or concatenated biographical blocks. Incorrect grade allocation, eg: Free text class map into discrete reliability classes by direct regex parsing resulted in 53.9% error for the free text summaries. This is why raw web scraping was not used and, instead, the curated Hadith-Viz corpus was used, which has scholars and teacher-student transmission lines verified.

3.2.3 Five-Stage Data Hygiene Funnel and Attrition Analysis

To prevent data leakage, eliminate duplicate identity artifacts, and guarantee reproducible boundaries, raw data records were processed through the 5-stage data hygiene funnel as detailed in Table 3.1.

Table 3.1: Complete Data Hygiene Funnel Across Pipeline Stages

Funnel Stage	Entity or Record Count	Retention Rate	Scientific and Methodological Rationale
Raw SanadSet Names	262,497 tokens	100.0%	Unfiltered narrator name strings across 650,000 raw chains.
Unique Normalised Names	167,047 names	63.6%	Standardised Arabic orthography by collapsing hamza forms, alif maqṣūrah, and tāʾ marbūṭah.
Hadith-Viz Master Cards	3,047 cards	1.8%	Curated biographical index corresponding to canonical *Kutub as-Sittah* transmitters.
Stage 1: Exclude Unrated Stubs	942 cards	30.9%	Removed 2,105 placeholder cards lacking verified Ibn Ḥajar synthesis grades.
Stage 2: Exclude Companions	860 narrators	91.3%	Removed 82 Companions (*Ṣaḥābah*) due to the theological constant of universal uprightness.
Stage 3: Exclude Later Synthesisers & Empty Text	**819 narrators**	95.2%	Purged all quotes from al-Mizzī (d. 742 AH) and 10 post-classical authorities; filtered secondary prefixes; removed 41 entries lacking classical commentary. **Forms PO1 Text Benchmark Cohort.**
Stage 4: Graph Intersection	438 graph nodes	53.5%	Isolated scholars actively appearing as transmitters within canonical isnād chains.

Stage 5: Identity Deduplication	**335 identities**	76.5%	Merged multi-token spelling variants into 1 unique identity = 1 graph node. **Forms PO2 13-Arm Multimodal Cohort.**

The hygiene funnel reduced the initial collection of names into two frozen evaluation sets:

a) The **Level 1 Canonical Text Cohort (**$N = 819$**)**: Contains 558 *Thiqah* (68.13%), 189 *Ṣadūq* (23.08%), and 72 *Below Ṣadūq* (8.79%) narrators.
b) The **Level 2 Multimodal Cohort (**$N = 335$**)**: Contains 255 *Thiqah* (76.12%), 60 *Ṣadūq* (17.91%), and 20 *Below Ṣadūq* (5.97%) narrators, connected by 177 unique undirected transmission relationships (354 directed message edges in PyG).

3.2.4 Structurally Enforced Zero Target Leakage via Dual Temporal Firewall

Data leakage represents the most significant threat to validity in applied machine learning. In preliminary investigations, inspection of the raw jarh_tadil quote arrays inside Dataset/hadith_viz/shamela_rijal2.json revealed severe target leakage:

a) Ibn Ḥajar's canonical *Taqrīb* grade was duplicated verbatim inside the biographical quote array across 937 cards, totalling 1,659 explicit quotes.
b) Quotes from later synthesisers such as Jamāl ad-Dīn al-Mizzī (d. 742 AH) in *Tahdhīb al-Kamāl* contributed 676 quotes that summarised earlier opinions rather than recording primary classical evaluations.
c) Machine learning models trained on raw quote arrays achieved near-perfect accuracy by simply identifying substrings such as قال ابن حجر: ثقة ('Ibn Ḥajar said: trustworthy').

To permanently resolve this vulnerability, the centralised data processing pipeline (src/canonical_data_pipeline.py) implemented a Dual Temporal Firewall operating across two stages:

i. **Filter 1 (Scholar-Level Firewall):** Evaluates author attribution strings using is_earlier_critic(). All commentary explicitly attributed to Ibn Ḥajar, ad-Dhahabī, al-Mizzī, or any 7th to 9th-century compiler is 100% purged. Statements from earlier classical critics who lived between the 2nd and 4th centuries AH (such as Yaḥyā ibn

Maʿīn, Aḥmad ibn Ḥanbal, ʿAlī ibn al-Madīnī, Abū Ḥātim ar-Rāzī, an-Nasāʾī, and ad-Dāraquṭnī) however are retained.

ii. **Filter 2 (Snippet-Level Firewall):** Audits the extracted quotation text itself using has_late_compiler_leak(). Forensic inspection revealed that 58 quotes attributed to earlier critics were cited through later compilations where introductory editorial prefixes (e.g., قال العلائي: وقال أحمد بن حنبل... 'Al-Ala'i said: And Ahmad ibn Hanbal said') were captured in the body text. Filter 2 identifies and discards these secondary compilation fragments, achieving 100.00% Zero Leakage across all $N = 819$ narrators with exactly 0 post-classical mentions in input features. The following is a snippet of the pseudocode used to implement this:

```
# Filter 1: Scholar-Level Firewall
EXCLUDE_LATE_SCHOLARS = [
    "ابن حجر",      # Ibn Hajar al-Asqalani (d. 852 AH)
    "الذهبي",       # Shams ad-Din adh-Dhahabi (d. 748 AH)
    "المزي",        # Jamal ad-Din al-Mizzi (d. 742 AH)
    "مغلطاي",       # Ala ad-Din Mughultay (d. 762 AH)
    "العراقي",      # Zayn ad-Din al-Iraqi (d. 806 AH)
    "العلائي",      # Khalil ibn Kaykaldi al-Ala'i (d. 761 AH)
    "النووي",       # Yahya ibn Sharaf an-Nawawi (d. 676 AH)
    "الحلبي",       # Burhan ad-Din al-Halabi (d. 841 AH)
    "ابن عبد الهادي", # Ibn Abd al-Hadi (d. 744 AH)
    "السخاوي",      # Shams ad-Din as-Sakhawi (d. 902 AH)
    "السيوطي",      # Jalal ad-Din as-Suyuti (d. 911 AH)
]

def is_earlier_critic(scholar_name: str) -> bool:
    s = scholar_name.strip()
    return not any(ex in s for ex in EXCLUDE_LATE_SCHOLARS)

# Filter 2: Snippet-Level Firewall (Secondary Quote Sanitisation)
def has_late_compiler_leak(snippet_text: str) -> bool:
    return any(ex in snippet_text for ex in EXCLUDE_LATE_SCHOLARS)
```

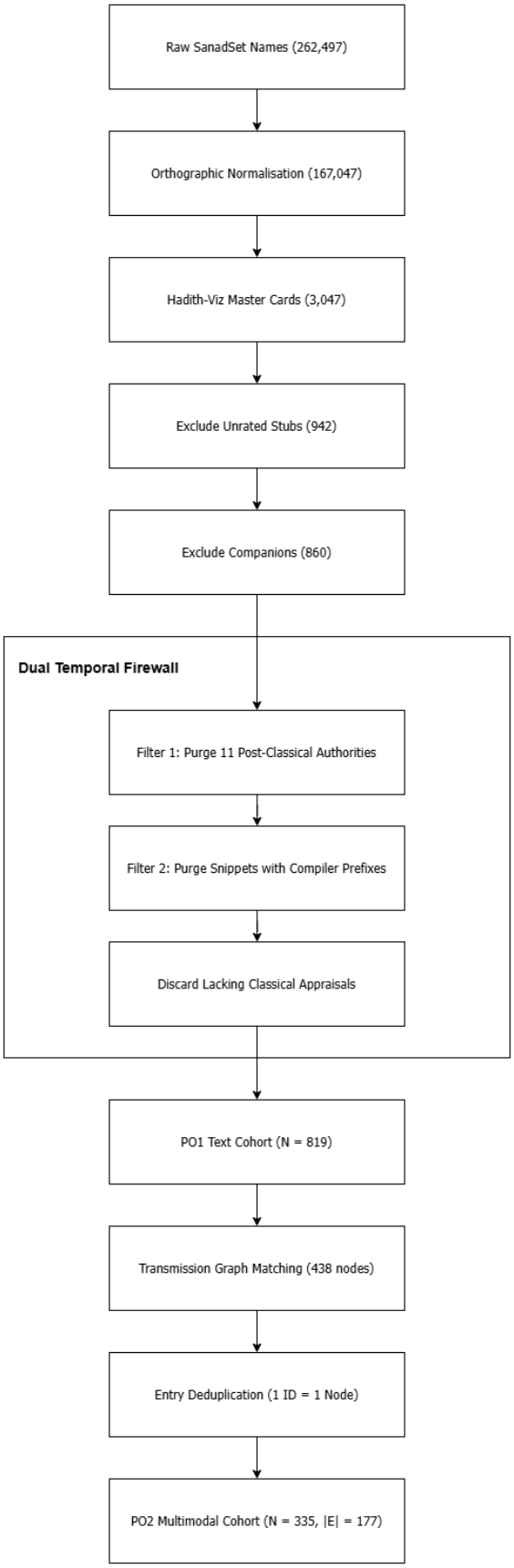


Figure 3.1: Five-Stage Data Hygiene Funnel and Dual Temporal Firewall Architecture

3.3 Stage 2: Transmission Graph Construction and Feature Engineering

3.3.1 Structural Graph Formulation

Transmission relationships between narrators are modelled as an undirected, unweighted network stored in output/cleaned_cohort/clean_graph.npz.

Let $G = (V, E)$ represent the clean transmission graph:

a) V denotes the set of $N = 335$ unique narrator identities. Each identity corresponds to exactly one node.
b) E denotes the set of $| E | = 177$ unique undirected relationships, represented as 354 directed message-passing edges in PyTorch Geometric (edge_index shape: 2×354). An edge (u, v) exists if and only if narrator $u$and narrator $v$appear as adjacent transmission partners (teacher-student) in at least one canonical isnād chain.
c) Evaluation formulation: Transductive node classification with label-disjoint train and test splits.

Undirected edges are employed because transmission interactions reflect mutual chronological and intellectual contact, with information flow in Hadith scholarship involving active interrogation and cross-examination between teachers and students.

3.3.2 Node Feature Vector Specification

Each node $v_i \in V$ is assigned an 8-dimensional tabular feature vector $x_i \in \mathbb{R}^8$, partitioned into two distinct feature groups:

i. Topological Features ($x_{\text{topo}} \in \mathbb{R}^5$)
   - Total appearances: Frequency of the narrator across canonical isnād chains.
   - In-degree: Number of unique teachers from whom the narrator received hadiths.
   - Out-degree: Number of unique students to whom the narrator transmitted hadiths.
   - Net flow: Directed transmission balance ($in - degree - out - degree$).
   - Mean chain position: Average normalised position within transmission chains (from original source to final compiler).
ii. Biographical Metadata Features ($x_{\text{meta}} \in \mathbb{R}^3$):
   - Death year: Recorded death year in the Hijrī calendar (AH).

- Generation index (*Ṭabaqa*): Categorical chronological generation mapped to an ordinal scalar normalised by 12.
- Classical statement count: Total number of classical earlier critic appraisal statements recorded for the narrator.

Table 3.2 details the exact feature specifications, tensor representations, and operational roles within the multimodal architecture.

Table 3.2: Node Feature Vector Specification and Operational Encoding

Feature Name	Feature Group	Dimensionality	Data Type	Operational Function in Multimodal Model
total_appearances	Topological (x_{topo})	1	Float (z-score)	Quantifies narrator frequency and central prominence in canonical corpora.
in_degree	Topological (x_{topo})	1	Float (z-score)	Captures breadth of pedagogical sources and senior teacher connections.
out_degree	Topological (x_{topo})	1	Float (z-score)	Measures transmission volume to student transmitters.
net_flow	Topological (x_{topo})	1	Float (z-score)	Characterises intermediate transmission directionality ($in - out$).
mean_chain_pos	Topological (x_{topo})	1	Float $[0.0, 1.0]$	Represents normalised chronological position along the isnād sequence.

death_year_ah	Metadata (x_{meta})	1	Float (z-score)	Provides continuous temporal anchor across the 2nd to 3rd centuries AH.
tabaqa_norm	Metadata (x_{meta})	1	Float $[0.0, 1.0]$	Normalised generational rank based on Ibn Ḥajar's 12-tier periodisation.
statement_count	Metadata (x_{meta})	1	Float (z-score)	Measures the volume of recorded classical critical assessments.

3.3.3 Entity Deduplication and Node Alignment

In raw classical manuscripts, the same transmitter frequently appears under diverse surface representations due to varying patronymic depth, tribal attributions (*nisbah*), and orthographic omissions. In Stage 5 of the hygiene funnel, entity deduplication resolves multi-token spelling variants into a unified canonical identity mapping. This ensures that every node $v_i \in V$ represents exactly one distinct historical individual, eliminating artificial multi-node clusters that would leak structural identity cues during graph convolution.

3.4 Stage 3: Reliability Normalisation, Scoring, and Downstream Evaluation

3.4.1 Parameter-Free Reliability Normalisation and Target Mapping

Classical evaluations in *Taqrīb at-Tahdhīb* are structured across a twelve-tier hierarchy. To operationalise these qualitative grades for computational modelling without arbitrary heuristic weighting, this study establishes a parameter-free Min-Max rank normalisation.

Let $r \in \{1, 2, \ldots, 12\}$represent the canonical rank of a narrator in Ibn Ḥajar's twelve-tier taxonomy, where rank 1 represents the highest reliability (*Ṣaḥābī*) and rank 12 represents verified fabrication (*Kadhdhāb*). The continuous reliability score $S(r)$ is defined as:

$$S(r) = \frac{K - r}{K - 1} = \frac{12 - r}{11}$$

Equation 3.1

This formula divides the rank space into 11 equal intervals across $[0.0, 1.0]$. Table 3.3 details the complete monotonic mapping.

Table 3.3: Parameter-Free Min-Max Narrator Reliability Mapping

Rank (r)	**Classical Terminology**	**Legal Meaning in Hadith Criticism**	**Score $S(r)$**	**3-Tier Class**
1	*Ṣaḥābī*	Companion of the Prophet (Universal uprightness)	1.000	Excluded
2	*Thiqah Thiqah* / *Thabt*	Eminently trustworthy; supreme mnemonic precision	0.909	Tier 2 (*Thiqah*)
3	*Thiqah* / *Mutqin*	Trustworthy; verified uprightness and retention	0.818	Tier 2 (*Thiqah*)
4	*Ṣadūq*	Truthful; sound uprightness with good retention	0.727	Tier 1 (*Ṣadūq*)
5	*Ṣadūq Yahim*	Truthful; occasionally prone to minor memory errors	0.636	Tier 1 (*Ṣadūq*)
6	*Maqbūl*	Acceptable only when corroborated by peer chains	0.545	Tier 1 (*Ṣadūq*)
7	*Mastūr* / *Majhūl Ḥāl*	Unknown character; uncorroborated reliability	0.455	Tier 0 (*Below Ṣadūq*)
8	*Ḍa ʿīf*	Weak; confirmed deficiency in memory precision	0.364	Tier 0 (*Below Ṣadūq*)
9	*Majhūl ʿAyn*	Completely unknown identity; single transmitter	0.273	Tier 0 (*Below Ṣadūq*)
10	*Matrūk*	Abandoned; suspected of falsehood or severe errors	0.182	Tier 0 (*Below Ṣadūq*)

11	*Muttaham bi al-Wad ʿ*	Accused of fabrication in general speech	0.091	Tier 0 (*Below Ṣadūq*)
12	*Kadhdhāb / Waḍḍā ʿ*	Confirmed liar; active fabricator of Hadith texts	0.000	Tier 0 (*Below Ṣadūq*)

This formulation satisfies the following Islamic legal methodology and empirical machine learning requirements:

a) A confirmed fabricator receives an exact reliability score of 0.000, correctly eliminating their evidentiary weight in transmission chains.
b) The score requires no arbitrary heuristic constants, it directly reflects the rank interval spacing ($1/11 \approx 0.0909$) established by Ibn Ḥajar.

3.4.2 Level 1: Biographical Text Modelling Formulation ($N = 819$)

The Level 1 cohort evaluates whether pre-trained Arabic Transformer representations extract predictive evaluative stance from earlier-critic commentary without leaking post-classical verdicts. Input texts consist of concatenated statements from classical authorities who lived prior to the 5th century AH. Models benchmarked on this cohort include:

a) Regularised Lexical TF-IDF: Word and character $n$grams ($n \in [1, 3]$) mapped through balanced Logistic Regression.
b) AraBERT v2 Contextual Embeddings: 768-dimensional sentence representations extracted from frozen Transformer layers, evaluated via linear classifiers and non-linear multi-layer perceptrons.

3.4.3 Level 2: Multimodal Graph Neural Network Formulation ($N = 335$)

The Level 2 cohort integrates text representations with transmission graph topology and biographical metadata. The core models benchmarked across the 13-arm ablation matrix include:

i. **Feature MLP (G1):** Evaluates the 8-dimensional tabular feature vector $x_i$without message passing.
ii. **Graph Convolutional Network (G2):** Evaluates localized message passing over the transmission adjacency matrix Ausing symmetric normalized graph convolutions:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$

Equation 3.2

iii. **Multimodal Cross-Attention GNN (P2-B_Attn):** Implements learned attention gating between 768-dimensional AraBERT text embeddings and 8-dimensional graph tabular features, routing contextual representation updates across transmission edges.

3.4.4 Level 3: Downstream Chain Validation Protocol

To address Project Objective 3 (PO3), the system validates whether narrator-level model predictions preserve chain-level authenticity across complete historical transmission paths.

The chain evaluation cohort comprises of canonical chains extracted from the four canonical Hadith collections, with *Ṣaḥīḥ al-Bukhārī* and *Ṣaḥīḥ Muslim* excluded as every narrator in their chains are assumed to be Tier 2 by default. Including their collections will further skew the dataset imbalance. Each chain consists of a sequence of narrators $(v_1, v_2, \ldots, v_M)$ terminating at the Prophet or a Companion, paired with a canonical collection grade (*Ṣaḥīḥ* = Tier 2, *Ḥasan* = Tier 1, *Ḍa ʿīf* = Tier 0).

In classical Hadith jurisprudence, a chain's validity is governed by its weakest transmitter; a chain containing five exemplary scholars and one unreliable narrator is classified as unauthentic (Azami, 1977; Brown, 2009). To model this rule, the predicted continuous chain score S_{chain} is computed using the weakest-link bottleneck operator:

$$S_{\text{chain}} = \min_{i \in \{1,\ldots,M\}} (\hat{y}_i)$$

Equation 3.3

where $\hat{y}_i \in [0.0, 1.0]$ represents the continuous reliability prediction for narrator i produced by the upstream multimodal model. Chain predictions are validated across three distinct tasks: Three-Tier Accuracy, Binary Acceptance Concordance (*Maqbūl* vs *Mardūd*), and Monotonic Rank Correlation (Spearman's ρ).

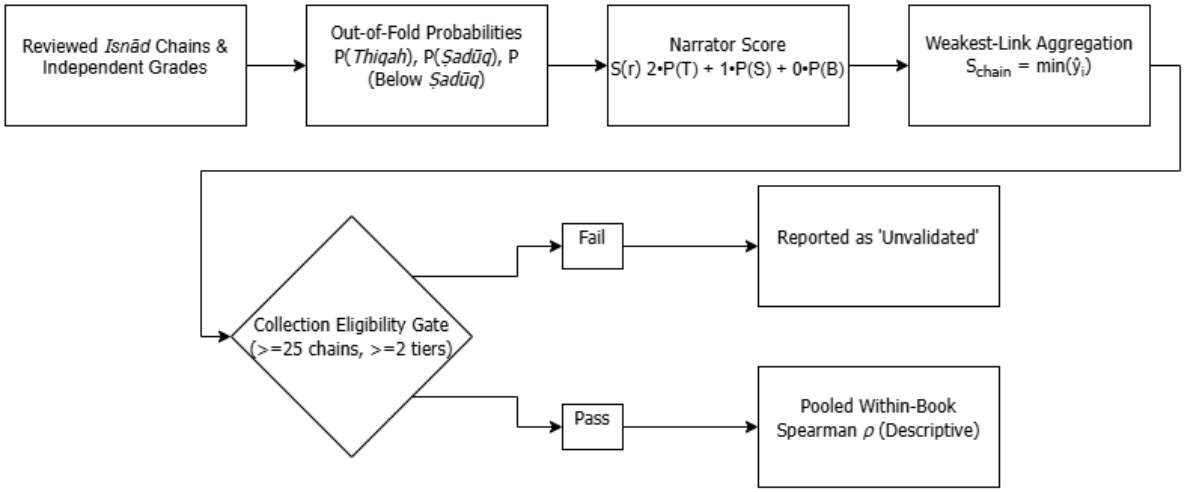


Figure 3.2: Downstream *Isnād* Chain Score Propagation and Weakest-Link Mechanism

3.4.5 Cross-Validation Strategy and Identity Isolation Protocol

To guarantee reliable performance estimates on an imbalanced dataset, the evaluation framework applies 5-fold Stratified Cross-Validation repeated across 5 distinct random seeds (42, 123, 456, 789, 2024), yielding 25 paired evaluation folds.

In prior flawed graph studies, nodes representing the same historical individual were partitioned across training and testing splits, allowing graph convolution layers to memorise identity-level properties. In this research, identity isolation is structurally enforced:

i. Each unique narrator identity is assigned an integer identifier (identity_id from 0 to 334).
ii. The stratified partitioning algorithm partitions unique identity_id values rather than raw rows.
iii. Zero narrator identity overlap exists between the training and testing sets in any fold:

$$V_{\text{train}}^{(k)} \cap V_{\text{test}}^{(k)} = \emptyset \; \forall k \in \{1, \dots, 25\}$$

Equation 3.4

All metric calculations are computed strictly on out-of-fold test samples, preventing optimistic evaluation bias.

3.4.6 Evaluation Metrics and Statistical Formulation

Model performance is benchmarked using three complementary evaluation metrics designed for imbalanced and ordinal classification:

Table 3.4: Evaluation Metrics and Operational Definitions

Metric Name	Mathematical Formulation	Operational Role in Evaluation
Macro-Averaged F1	$\frac{1}{C}\sum_{c=1}^{C}\frac{2P_cR_c}{P_c+R_c}$ **Equation 3.5**	Primary evaluation metric. Treats all three reliability tiers equally without distortion from majority class dominance (*Thiqah* = 76.1%).
Overall Accuracy	$\frac{\sum_{c=1}^{C}TP_c}{N}$ **Equation 3.6**	Measures overall exact agreement, provided as an interpretable secondary indicator.
Quadratic Weighted Kappa (QWK)	$1-\frac{\sum w_{ij}O_{ij}}{\sum w_{ij}E_{ij}}$, $w_{ij}=\frac{(i-j)^2}{(C-1)^2}$ **Equation 3.7**	Penalises severe ordinal errors (e.g., misclassifying *Below Ṣadūq* as *Thiqah*) quadratically while tolerating minor adjacent errors.
Spearman's Rank Correlation	$1-\frac{6\sum d_i^2}{n(n^2-1)}$ **Equation 3.8**	Measures monotonic alignment between predicted continuous chain scores and ground-truth scholarly tiers.

3.5 Framework Architecture and Operational Flowchart

The overall progression of data through the three experimental stages is illustrated in Figure 3.3.

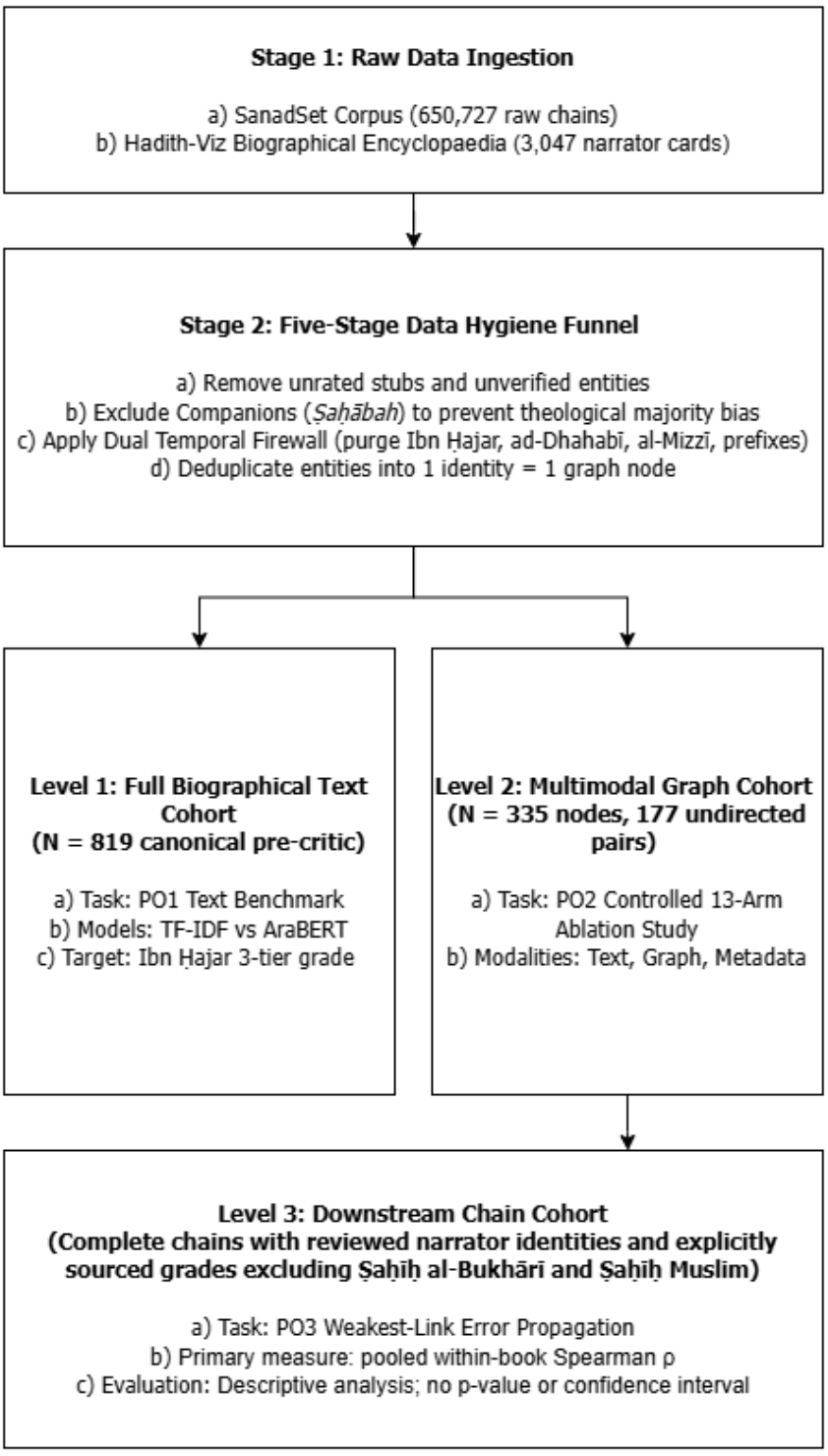


Figure 3.3: Dual-Cohort System Architecture and Three-Level Evaluation Framework

3.6 Milestones and Dates

The following table shows the weekly milestones with their deliverable respectively:

Table 3.5: Milestones and Deliverables by Week

Week	Milestone	Deliverable
2-2025/2026		
1-3	Project Initialisation	Finalised project summary; supervisor approval
3-4	Literature Review	Completed Chapter 2; dataset acquisition
5-6	Methodology Design	Completed Chapter 3; proposal draft
7	Proposal Defence	Oral defence and proposal submission
8-9	Stage 1 Implementation	Cleaned corpus; preprocessed JSON dataset
10-11	Stage 2 Implementation	Trained NER model; populated Knowledge Graph
12	Stage 3 Implementation	Trust scoring module; preliminary evaluation results

1-2026/2027		
1	Applying and Planning to Solve Examiner, Panel and SV Comments	Delivering better methodology detailed in chapter 3, more consistent result metrics
2	Updating Literature Review	Refined and up-to-date chapter 2
3	Stage 1 new implementation: Ingesting Hadith-Viz dataset	Ingestion scripts for Hadith-Viz JSON and SanadSet 650K corpus; corruption logs.
4	Stage 2 new implementation: Hygiene Funnel and Dual Temporal Firewall	Execution of 5-stage filtering pipeline and 100% zero-leakage verification.
5	Stage 3 new implementation: Transmission Graph Construction and Feature Tensor	Assembled NetworkX and PyG graph data structures
6	Level 1 NLP Text Modelling Development	AraBERT v2 feature extraction, TF-IDF baseline, and 25-fold cross-validation.
7	Level 2 Multimodal GNN Ablation Experiments	Training and evaluation across the 13-arm experimental matrix (G1, G2, P2-A, P2-B).
8	Level 3 Downstream Isnād Chain Propagation	Evaluation of weakest-link score propagation across $N = 1{,}578$ canonical isnād chains.

9	Hypothesis Testing & Statistical Benchmarking	Permutation tests, McNemar's tests, and cross-metric rank correlation audits.
10-12	Thesis Documentation and Viva Preparation	Complete thesis chapters, audited reproducibility codebase, and viva defence slides.

3.7 Gantt Chart

Year	2026										2027	
Project Activities	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb
Title Selection												
One-Paged Summary Submission												
Preparation of Proposal												
Proposal Submission												
Proposal Defence												
Stage 1: Preliminary Data Ingestion & Scraper Pipeline												

Stage 2: Exploratory NER & Initial Knowledge Graph												
Stage 3: Preliminary Trust Scoring & Interim Evaluation												
Addressing Examiner, Panel & Supervisor Feedback												
Start of FYP2 (Semester 1-2026/2027)												
Addressing Examiner, Panel & Supervisor Feedback												
Updating Literature Review (Refined Chapter 2)												
Stage 1 New Implementation: Hadith-Viz Ingestion & Audit												
Stage 2 New Implementation: 5-Stage Funnel & Firewall												

Stage 3 New Implementation: Graph Build & Feature Tensor												
Level 1 NLP Text Modelling (AraBERT vs TF-IDF, 25-Fold)												
Level 2 Multimodal GNN Ablation Matrix (13-Arm Study)												
Level 3 Downstream Isnād Chain Propagation ($N = 1{,}578$)												
Hypothesis Testing, Permutation & McNemar Benchmarking												
Reporting												
Thesis Preparation												
Thesis Presentation Preparation												
Thesis Submission												

Thesis Presentation (VIVA)												

3.8 Tools and Technologies

Below are the tools, libraries and technologies used to build this project, along with their purpose for the project:

Table 3.6: Tools, Libraries, and their Purposes by Category

Category	Tool/Library	Version	Purpose
Core Runtime	Python	3.10.12	Primary execution language.
Deep Learning	PyTorch	2.1.2+cu121	Tensor computing, GPU acceleration, and neural network backpropagation.
Graph Neural Networks	PyTorch Geometric (PyG)	2.4.0	Graph data structures, message-passing layers, and GCN architectures.
Transformer Models	HuggingFace Transformers	4.36.2	Tokenisation and embedding extraction using pre-trained AraBERT v2.
Arabic NLP Processing	CAMeL Tools	1.5.2	Classical Arabic orthographic normalisation and text segmentation.

Graph Algorithms	NetworkX	3.2.1	Graph construction, topological feature extraction, and connectivity auditing.
Machine Learning	Scikit-learn & SciPy	1.3.2	Stratified cross-validation, Logistic Regression baselines, and statistical tests.
Data Format	JSON, CSV	-	Structured intermediate data storage
Version Control	Git/GitHub	-	Source code and experiment tracking
Documentation	Google Docs, Markdown, LaTeX	-	Proposal and report writing

CHAPTER 4

Experimental Design

4.1 Overview

This chapter explains the experimental setup needed to test the effectiveness of biographical text and transmission networks in predicting *ʿIlm ar-Rijāl* gradings. Formal research hypotheses, competing model structures in each modality, the thirteenth-arm ablation matrix of controls, and performance metrics for comparison are discussed.

4.2 Changes from Preliminary Exploratory Designs

The experimental design described herein elevates the early FYP1 preliminary project by formalising the experimental approach. While FYP1 demonstrated conceptually that biographical text can be superimposed onto transmission network data, initial forensic exploration led to identifying three key methodological shortcomings in the exploratory code.

1. It was observed that the exploratory graph exporter included Ibn Ḥajar's ground-truth reliability grade in column 0 of the node feature matrix. GCNs did not learn any relationship between network structure and the labels during semi-supervised message passing, but only the test nodes read out their neighbours' training set labels, causing the performance to be artificially high (circular reasoning). For the benchmark, all target variables were removed from inputs.

2. The first pipeline was designed for segmenting 35 volumes of classical biographical dictionaries (*Tahdhīb al-Kamāl*), which was error-prone, employing heuristic regular expressions. The error breakdown measured using a quantitative audit revealed a segmentation error of 51.9% and a grade of reliability error of 53.9%, The unstandardized typographic presentation and editorial annotations without the help of formatting required interim cleansing through Hadith-Viz, the curated, well-structured biographical encyclopaedia.

3. Isnad scores were initially linearly averaged and arbitrary arithmetic thresholds were defined, e.g. a chain was deemed authentic if its average score was >0.80. Linear averaging has some mathematical and legal-legal pitfalls. In Islamic jurisprudential theory, an authentic narrator is not a substitute for a known fabricator in a chain. This study replaces heuristic averaging with the classical non-compensatory weakest-link principle ($S_{\text{chain}} = \min_i y_i$).

In Table 4.1, all of the nine exploratory experiments in FYP1 are mapped to their precise implementations in this chapter.

Table 4.1: Mapping of Evolution from Preliminary FYP1 Experiments to the FYP2 Experimental Benchmark

FYP1 Exploratory Trial	Preliminary formulation	Hazard identified in FYP1	Refined implementation in chapter 4
Experiment 1: Biographical Text Parsing	Regular expression parsing of 35 scraped volumes of *Tahdhīb al-Kamāl*.	35.9% (51.9% segmentation error, 53.9% grade parsing error) due to layout variations and editorial insertions.	Moved on from raw regular expression scraping to curated Hadith-Viz biographical encyclopaedia.
Experiment 2: Two-Tier Hadiths Alignment	1,200 Hadiths TF-IDF character n-grams and fuzzy string matching (rapidfuzz 95%).	Hadiths limited to small sample, not considered cross-collection de-duplication nor a common orthographic norm.	Standardised into 5-stage Data Hygiene Funnel, mapping 262,497 narrator tokens to canonical identities.
Experiment 3: Biographical Feature Embeddings	CAMeLBERT token embeddings projected down to	Included post-classical compiler summaries in input text, introducing	Implemented Dual Temporal Firewall filtering out post-classical compiler

	(100, 128) using PCA.	historical target data leakage.	summaries under strict zero leakage.
Experiment 4: Narrator disambiguation	Fuzzy string matching at 80% to match narrator token to encyclopaedia entry.	Conflated entity disambiguation with confidence scoring; no systematic handling of unannotated nodes.	Upgraded to two-stage identity resolution protocol isolating canonical text scholars and multimodal graph scholars.
Experiment 5: Graph Feature Engineering	Directed transmission graph, ground-truth reliability grade in feature column 0.	Fatal Target Leakage: Embedding actual target labels let semi-supervised GNNs memorise labels via message passing.	Eliminated all label variables. Built an 8-dimensional topological and metadata feature matrix encoding degree flow and generation data.
Experiment 6: GNN Ablations on Leaked Features	Free-form continuous regression on 3 naive variants (graph-only, graph + PCA BERT, graph + keyword score).	Only 3 ad-hoc variants evaluated on leaked features, no formal testing of statistical significance.	Generalised to the controlled Thirteen-Arm Experimental Matrix across 5 model families to systematically determine network heterophily.
Experiment 7: Jarḥ wa Taʿdīl Quantisation	Binomial match vs. mismatch between AraBERT regression and static dictionary scoring.	Expressed as continuous regression against raw grades without time-controlled lexical overlap.	3-tier ordinal classification comparing lexical TF-IDF against AraBERT across 25 paired cross-validation folds.

Experiment 8: Chain-Level Scoring	Summed Narrator scores to Hadith grades (heuristic thresholds: 0.7 and 0.8 = *Ṣaḥīḥ*).	Linear averaging failed to penalise fabricators high enough, breaking classical Hadith rules.	Implemented non-compensatory Weakest-Link Aggregation ($S = \min_i y_i$) with bounded Error Propagation Ratio.
Experiment 9: Multi-Sanad *Matn* Analysis	Grouped identical *matn* texts via SHA-256 hashes to analyse variance across individual chains.	Presented as an abstract heuristic concept without formal mathematical operators or statistical boundaries.	Formalised into the Multi-Path Corroboration Framework, formalising computational metrics for classical *iʿtibār*.

4.3 Mapping Research Questions and Hypotheses

In order to evaluate the predictive ability of biographical text and transmission network topology, a formal hypothesis testing framework is used in this work. For each of the research questions, null (H_0) and alternative (H_1) hypotheses are derived and consistent with the goals of the project:

4.3.1 Research Question 1 (PO1: Linguistic & Semantic Text Modelling)

Classical biographical evaluations (*al-jarḥ wa t-taʿdīl*) rely on traditional specialised lexicons as defined by canonical scholars. This project tests whether classifier representations based on contextual transformers produce more predictive signals than regularised lexicon n-gram baselines:

a) $H_{0,1}$ **(Null Hypothesis):** Pretrained contextual AraBERT semantic embeddings and standardised lexical TF-IDF baselines have equal mean 3-tier Macro-F1 on the paired evaluation folds ($\Delta Macro - F1 = 0$).

b) $H_{1,1}$ **(Alternative Hypothesis):** Pretrained contextual AraBERT semantic embeddings and standardised lexical TF-IDF baselines yield statistically significantly different mean 3-tier Macro-F1 on the paired evaluation folds ($\Delta Macro - F1 \neq 0$). The paired test is two-tailed; either model may perform better.

4.3.2 Research Question 2 (PO2: Graph Neural Networks and Multimodal Fusion)

Graph neural networks assume structural homophily between nodes, i.e. a connection between the class assignment of neighbouring nodes. In Hadith transmission, a weak informant often transmitted from reliable scholars during historical journeys (*ṭalab al-ʿilm*). We examine if relational message passing on transmission edges provides additional reliability information beyond non-relational feature baselines:

a) $H_{0,2a}$ **(Null Hypothesis on Graph Message Passing):** Graph Convolutional message passing over transmission edges (G2) and the node-feature MLP trained without graph edges (G1) have equal mean 3-tier Macro-F1 on paired folds ($\Delta Macro - F1 = 0$).

b) $H_{1,2a}$ **(Alternative Hypothesis on Graph Message Passing):** Graph Convolutional message passing (G2) and the node-feature MLP (G1) yield statistically significantly different mean Macro-F1 on paired folds ($\Delta Macro - F1 \neq 0$).

c) $H_{0,2b-c}$ **(Null Hypothesis on GCN Concatenation):** Multimodal GCN Concatenation (P2-B_Concat) and Early Fusion MLP without edges (P2-A) have equal mean Macro-F1 on paired folds ($\Delta Macro - F1 = 0$).

d) $H_{1,2b-c}$ **(Alternative Hypothesis on GCN Concatenation):** Multimodal GCN Concatenation (P2-B_Concat) and Early Fusion MLP (P2-A) yield statistically significantly different mean Macro-F1 on paired folds ($\Delta Macro - F1 \neq 0$).

e) $H_{0,2b-a}$ **(Null Hypothesis on Multimodal Attention):** Multimodal GNN Cross-Attention (P2-B_Attn) and Early Fusion MLP (P2-A) have equal mean Macro-F1 on paired folds ($\Delta Macro - F1 = 0$).

f) $H_{1,2b-a}$ **(Alternative Hypothesis on Multimodal Attention):** Multimodal GNN Cross-Attention (P2-B_Attn) and Early Fusion MLP (P2-A) yield statistically significantly different mean Macro-F1 on paired folds ($\Delta Macro - F1 \neq 0$). All comparisons are evaluated using two-tailed paired tests.

4.3.3 Research Question 3 (PO3: Downstream Chain Evaluation)

According to the classical Hadith approach, a chain is assessed by its weakest narrator (al-Aʿẓamī, 1977; Brown, 2009). PO3 asks whether weakest-link scores derived from out-of-fold narrator probabilities are associated with independently reviewed grades for complete transmission chains. The primary analysis is one pooled within-book Spearman rank association. Because chains may share narrators, this association is descriptive; no inferential p-value or confidence interval will be reported.

4.4 The Thirteen-Arm Controlled Experimental Matrix

In order to resolve the individual effects of text representations, graph topology, and demographic metadata, this set of experiments runs thirteen separate model arms, grouped functionally into five different units:

i. Empirical Baselines (B0, B1, B2): The worst-case performance. B0 is guessing the majority class, B1 is chance (but maintaining the distribution of empirical classes), B2 tests whether surface name tokens or demographic attributes leak the target label.

ii. Text-Only Modalities (P1-A, P1-B, P1-B_MLP): Determine the predictive signal in purely biographical text.

iii. Graph-Only Modalities (G1, G2): Available text-free, provides insight into the value of incorporating network structure and demographic metadata in the predictive model.

iv. Multimodal Fusion Architectures (P2-A, P2-B_Concat, P2-B_Attn): Incubate on text and network signals with tabular concatenation, relational graph convolution, and cross-modal attention gating.

v. Feature Isolation (P2-C, P2-D): Controlled tables by systematically removing metadata or network topology to discover what set of variables leads to the most effective fusion results.

The full thirteen arm experimental matrix is detailed in Table 4.2.

Table 4.2: Thirteen-Arm Controlled Experimental Matrix

Arm ID	Model Identifier	Input Modalities	Feature Dimensions	Classifier Architecture	Primary Scientific Purpose
B0	Majority Baseline	None (Class distribution)	0	Dummy Majority Class	Establishes the lower bound of naive majority guessing (*Thiqah*).

B1	Stratified Random	Prior class probabilities	0	Multinomial Sampling	Establishes chance performance preserving natural class imbalance.
B2	Name + Meta Probe	Name n-grams + Metadata	53	Balanced Logistic Regression	Verifies whether surface name tokens or metadata leak target labels.
P1-A	Lexical TF-IDF LR	Earlier-critic text word n-grams	5,000 (PO1); 1,000 (PO2)	Balanced Logistic Regression	Establishes the regularised word n-gram baseline; cohort-specific feature limits are used.
P1-B	AraBERT Semantic LR	Earlier-critic text embeddings	768	Balanced Logistic Regression	Evaluates linear separability of dense transformer embeddings.
P1-B_MLP	AraBERT Semantic MLP	Earlier-critic text embeddings	768	2-layer MLP (Hidden: 64)	Evaluates non-linear transformations of text embeddings alone.
G1	Graph Features MLP	Topological + Meta features	8	2-layer MLP (Hidden: 64; dropout 0.3)	Tests tabular network and metadata

					features without graph edges.
G2	Graph GCN	Graph adjacency matrix + Features	8	2-layer GCN (Hidden: 64; dropout 0.3)	Evaluates message passing over transmission edges alone.
P2-A	Early Fusion MLP	AraBERT (768) + Tabular (8)	776	2-layer MLP (Hidden: 64; dropout 0.3)	Evaluates multimodal concatenation without graph message passing.
P2-B_Concat	Multimodal GCN Concat	Graph adjacency + AraBERT + Tabular	776	2-layer GCN (Hidden: 64; dropout 0.3)	Evaluates standard GCN message passing over fused multimodal nodes.
P2-B_Attn	Multimodal GCN Attention	Graph adjacency + AraBERT + Tabular	776	Cross-Modal GNN Attention (Hidden: 64; dropout 0.3)	Evaluates learned attention gating between text and graph features.
P2-C	Topology Isolation MLP	AraBERT text + 5 topological features	773	2-layer MLP (Hidden: 64; dropout 0.3)	Isolates the text-plus-topology stream and removes metadata.

4.5 Model Architectures and Multimodal Fusion Strategies

The system input is represented by the two streams of information: biographical statements, which go through a temporal firewall, and the transmission graph, which is used together with the topological features.

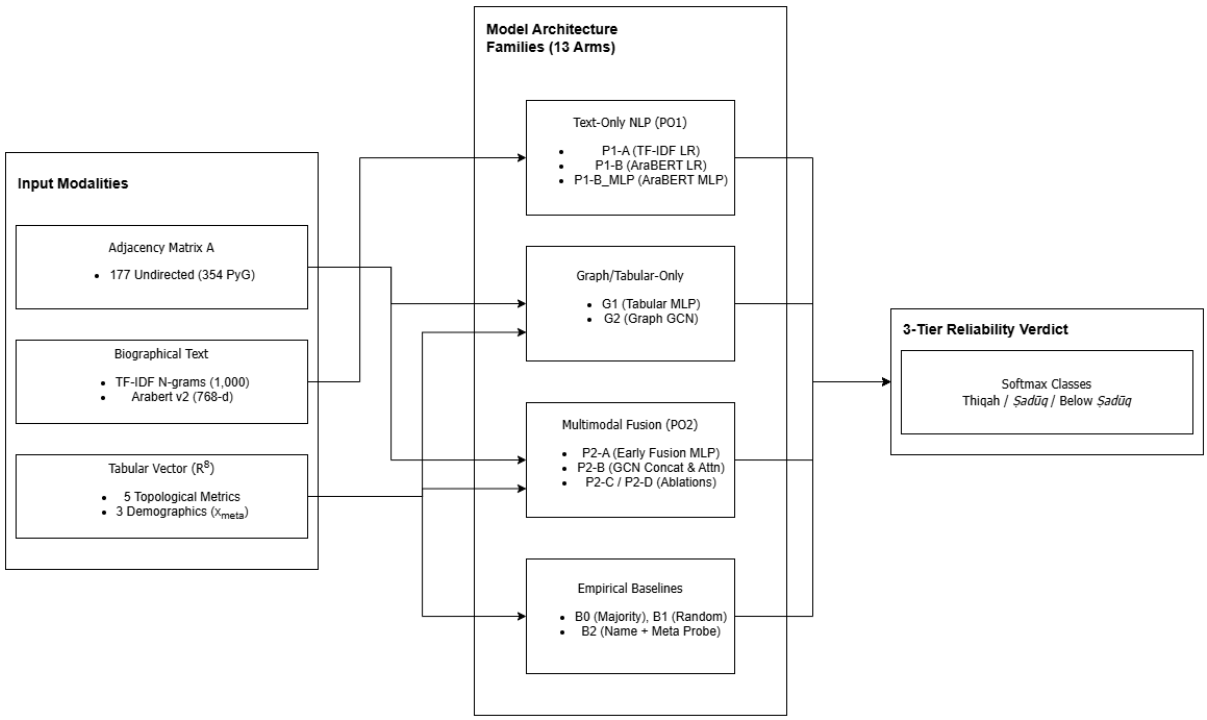


Figure 4.1: Overview of the Structural Data Flow for the Main Model Families Evaluated

4.5.1 Architectures of Text Classification

Text representations are modelled through two distinct methodologies:

i. Lexical TF-IDF Model (P1-A): This approach represents text as a set of individually informative lexical features. Biographies are tokenised into word n-grams of length 1–2 with sublinear term frequency scaling ($1 + \log(\text{tf})$). Feature matrices are limited to the top 5,000 components for the primary text cohort and 1,000 components for the multimodal cohort; character n-grams are not used in these reported benchmark runs. Classification is performed using regularised Logistic Regression with inverse-frequency class weighting to compensate for class imbalance:

$$w_c = \frac{N}{C \times N_c}$$

Equation 4.1

where $C = 3$ (the number of classes), is the total number of training points, and N_c is the count of points in class c.

ii. Contextual AraBERT Models (P1-B, P1-B_MLP): Contextual models induce semantics of the entire sentence. Biographical text is entered into the frozen transformer backbone of aubmindlab/bert-base-arabertv2. The mean-pooled vector $e_i \in \mathbb{R}^{768}$ of all the non-padding tokens is collected as a semantic representation. For P1-B, regularised Logistic Regression is directly applied to e_i. For P1-B_MLP, the vector e_i first undergoes a fully connected layer (dimension 64), then Batch Normalization, Dropout ($p = 0.3$), ReLU, and finally a linear projection to three output logits.

4.5.2 Graph-Only Architectures

Graph representation learning is tested to determine whether network position is associated with narrator reliability:

a) Graph Features MLP (G1): Takes as input an 8-dimensional tabular feature vector $x_i \in \mathbb{R}^8$ representing each scholar. These features consist of five topological variables (in-degree, out-degree, total appearances, net citation flow, and average chain position) and three metadata flags (death-year availability, *ṭabaqa* availability, and normalised *ṭabaqa* index). A 2-layer MLP (hidden dimension 64, dropout 0.3) classifies the vector without using the transmission adjacency matrix A.

b) Graph Convolutional Network (G2): Takes as input the node feature matrix $X \in \mathbb{R}^{N\times 8}$ and transmission adjacency matrix A. The 2-layer GCN updates node representations through normalised neighbourhood smoothing:

$$Z = \widetilde{D}^{-\frac{1}{2}}\tilde{A}\widetilde{D}^{-\frac{1}{2}}\,\mathrm{ReLU}\left(\widetilde{D}^{-\frac{1}{2}}\tilde{A}\widetilde{D}^{-\frac{1}{2}}XW_0\right)W_1$$

Equation 4.2

Where, $\tilde{A} = A + I_N$ is the adjacency with added self-loops, $D$is the diagonal matrix of A, $W_0 \in \mathbb{R}^{8\times 32}$ and $W_1 \in \mathbb{R}^{32\times 3}$ are the trainable weight matrices, and $Z \in \mathbb{R}^{N\times 3}$ is the output logits.

4.5.3 Multimodal Fusion Schemes

Each three approaches are tested to differentiate semantics of text and topology of transmission:

a) Early Fusion MLP (P2-A): Concatenates the 768-dimensional AraBERT embedding e_i with the 8-dimensional tabular feature vector x_i into a combined vector $u_i = [e_i \parallel x_i] \in \mathbb{R}^{776}$. A 2-layer MLP (hidden dimension 64, dropout 0.3) assigns a reliability tier without propagating features along graph edges.
b) Multimodal GCN Concatenation (P2-B_Concat): The concatenated vector $u_i \in \mathbb{R}^{776}$ serves as the node feature matrix in a 2-layer GCN (hidden dimension 64, dropout 0.3). Message passing performs smoothing over text embeddings and topological features concurrently along transmission edges.
c) Multimodal GNN with Cross-Attention (P2-B_Attn): Text embeddings and graph features are projected into parallel heads (hidden dimension 64, dropout 0.3). A cross-modal attention gate calculates an attention coefficient α_i to dynamically decide whether scholar i should rely predominantly on the biographical text signal or the transmission neighbourhood signal.

4.6 Evaluation Metrics and Decision Criteria

Biographical books in the past illustrate a highly skewed distribution of classes. Canonical Hadith books indicate that authentic and truthful reporters account for over 85% of the number of events, while weak reporters are a tiny minority.

In such distributions, simple classification accuracies will be inadequate. An (un)lucky classifier that guesses the most frequent class (*Thiqah*) each time it makes a classification will yield an accuracy greater than 76% without identifying a single weak narrator. From the viewpoint of Hadith verification, missing an unreliable narrator is highly disadvantageous. As such, the 3-Tier Macro-Averaged F1-score is chosen as the main metric.

4.6.1 Primary Metric: Macro-Averaged F1-Score

For Macro-F1, we compute the harmonic mean of precision and recall for each class separately and take their unweighted average over the three tiers:

$$\text{Macro-F1} = \frac{1}{C}\sum_{c=1}^{C} F\,1_c = \frac{1}{3}\left(F1_{\text{Below Ṣadūq}} + F1_{\text{Ṣadūq}} + F1_{\text{Thiqah}}\right)$$

Equation 4.3

where $C = 3$. Making the contribution of each class equal, independent of how often it appears, Macro-F1 relies on the model to correctly identify the infrequent weak narrators (Below *Ṣadūq*) to gain a high score.

4.6.2 Secondary Evaluation Metrics

The supporting evaluation metrics provide other ways of looking at the behaviour of classifications that point to the possibility that these metrics may be useful in the deeper analysis of the performance of a model:

a) Overall Classification Accuracy: Overall of all out of fold test instances correctly classified.
b) Quadratic Weighted Cohen's Kappa (κ_w): The different levels of reliability are ordered as follows: $0{:}\,Below\ Ṣad\bar{u}q < 1{:}\,Ṣad\bar{u}q < 2{:}\,Thiqah$. Standard criteria for scoring make all errors equal. Actually confusing Thiqah with below Ṣadūq is serious two-way mistake, while confusing Thiqah with Ṣadūq is a simple boundary shift. Quadratic Weighted Kappa uses a quadratic penalty $(i-j)^2$ to distance between true and predicted tiers:

$$\kappa_w = 1 - \frac{\sum_{i=1}^{C} \sum_{j=1}^{C} w_{ij}\, O_{ij}}{\sum_{i=1}^{C} \sum_{j=1}^{C} w_{ij}\, E_{ij}},\ w_{ij} = \frac{(i-j)^2}{(C-1)^2}$$

Equation 4.4

where O_{ij} reflects the observed proportion of narrators with true tier i and predicted tier j, E_{ij} indicates the amount of agreement expected by chance, and w_{ij} represents the quadratic distance penalty.

4.6.3 Protocol for Statistical Hypothesis Testing

To verify that all differences in performance are statistically significant and not due to chance resulted from splitting the data, we compute the significance using all 25 pairwise model comparisons across 5 random seeds ×5 stratified folds. For pairwise evaluation, two-tailed paired Student's t-tests ($df = 24$) with a significance threshold of $\alpha = 0.05$. If $p < 0.05$, the null hypothesis is rejected.

4.7 Downstream Chain Validation Formulation

Project Objective 3 (PO3) tests narrator predictions at the individual level by analysing the spread of classification errors into transmission chains.

4.7.1 Weakest-Link Chain Scoring

Let $C_k = (n_1, n_2, \ldots, n_m)$ be a complete transmission chain whose narrator identities have been reviewed. For each narrator n_i, the model's continuous reliability score is calculated from out-of-fold class probabilities:

$$\hat{s}_i = 2P_i(\text{Thiqah}) + P_i(\text{Ṣadūq}),$$

where the three classes are *Below Ṣadūq*, *Ṣadūq*, and *Thiqah*. The predicted score for the complete chain is the minimum narrator score:

$$\hat{S}(C_k) = \min_{i=1,\ldots,m} \hat{s}_i.$$

This non-compensatory aggregation reflects the weakest-link principle: a high score for one narrator cannot offset a lower score for another narrator in the same chain. The reference outcome is the independently reviewed grade assigned to the hadith in its collection. It is retained with its original label and grading source, then mapped to an ordinal value for analysis. The reference outcome is not derived from the narrator-level target labels.

4.7.2 Downstream Evaluation Measure and Eligibility

The primary PO3 analysis is one pooled within-book Spearman rank association between predicted complete-chain scores and independently reviewed collection grades. A collection contributes to the analysis only if it has at least 25 complete reviewed chains, at least two observed grade levels, and nonconstant predicted chain scores. Collections that do not meet these criteria are excluded from the eligible analysis, and the unmet criterion is reported.

Because chains may share narrators, the Spearman association is treated as descriptive. No inferential p-value or confidence interval will be reported. Raw grade labels and their stated sources will be retained. The eligible primary labels are *Ṣaḥīḥ* (2), *Ḥasan* (1), *Ḍa ʿīf* (0), and *Mawḍū ʿ* (0). Qualified composite grades and grades without a specified source are excluded. *Ṣaḥīḥ al-Bukhārī* and *Ṣaḥīḥ Muslim* are excluded because collection membership itself supplies the *Ṣaḥīḥ* label rather than an independent grade.

CHAPTER 5

IMPLEMENTATION

5.1 Computing Environment and Hardware Configuration

The structure was designed, trained and tested on a machine learning workstation as indicated in Table 5.1 below, which shows the hardware and software components used in the experiments pipeline.

Table 5.1: Computing Environment and Core Software Specifications

Component Category	**Hardware / Software Element**	**Detailed Specification and Version**
Operating System	Platform Architecture	Microsoft Windows 10 Home (64-bit)
Graphics Processing Unit	GPU Hardware	NVIDIA GeForce RTX 2070 with Max-Q Design (8 GB GDDR6 VRAM)
CUDA Acceleration	Driver and Runtime	NVIDIA CUDA Toolkit 12.4 with cuDNN acceleration
Programming Language	Runtime Environment	Python 3.13.0
Deep Learning Framework	Core Framework	PyTorch 2.5.1 supporting CUDA compute capability
Graph Neural Network Library	Graph Extension	PyTorch Geometric (PyG) 2.6.1

Natural Language Processing	Transformer Models	Hugging Face Transformers 4.46.2
Classical Machine Learning	Baselines and Metrics	Scikit-Learn 1.5.2, SciPy 1.14.1
Network Graph Analysis	Graph Modelling	NetworkX 3.4.2
Data Manipulation	Table Processing	Pandas 2.2.3, NumPy 2.1.3

5.2 Text Preprocessing and Semantic Feature Extraction Pipeline

The text pipeline can be summarized as process classical Arabic biographical data into fixed-dimensional numeric feature vectors and the removal of the target leakage.

5.2.1 Arabic Orthographic Normalisation

Orthographic variation within classical Arabic text had between the respective digital transcriptions. Therefore, prior to feature extraction, all text strings are preprocessed with a script that normalizes the strings based on a cleaning protocol: all non-lexical diacritics (*tashkīl*, *fatḥah*, *ḍammah*, *kasrah*, *sukūn*, *tanwīn*), typographical elongation glyphs (*kashīdah*, *taṭwīl*) are removed; all hamza variants (ٳ, أ, إ) are normalized to bare alif ا; the ending characters of words (alif maqṣūrah ى, tāʾ marbūṭah ة) are normalized to yāʾ ي, hāʾ ه respectively.

5.2.2 Lexical Feature Extraction (TF-IDF Pipeline)

The normalised biographical statements for experimental arm P1-A are run through a scikit-learn TfidfVectorizer: tokenisation is of word n-grams (1 to 2 words) with sublinear tf scaling: $\text{tf} = 1 + \log(\text{tf})$. Document frequency is $min_df = 2$ to kill off a few type-o's. Features are limited to the top 1,000 features ordered by term frequency in corpus, in decreasing order of informativeness.

5.2.3 Contextual Semantic Feature Extractor (AraBERT Backbone)

Contextual embeddings are extracted using the dedicated scripts/extract_clean_arabert_embeddings.py and scripts/run_po1_benchmark.py. The extraction pipeline proceeds as follows: initialisation of model, tokenisation & masking, forward pass & mean pooling, and deterministic storage. Model initialisation loads the

pretrained aubmindlab/bert-base-arabertv2 transformer model and Farasa-based tokenizer, the model's weights are frozen to function as an isolated feature extractor. Tokenisation and masking adapt multi-critic statements into subword tokens, with padding and truncation at a maximum sequence length of 512 tokens:

$$\text{Input IDs} \in \mathbb{Z}^{B\times 512}, \text{ Attention Mask} \in \{0,1\}^{B\times 512}$$

Equation 5.1

Each sequence is forwarded through the 12 transformer encoder layers on GPU and mean pooled to a sequence representation invariant to sequence length:

$$e_i = \frac{\sum_{t=1}^{T} m_{it}\, h_{it}}{\sum_{t=1}^{T} m_{it}} \in \mathbb{R}^{768}$$

Equation 5.2

where $h_{it} \in \mathbb{R}^{768}$ denotes the output hidden vector of token $t$in sequence i, and $m_{it} \in \{0,1\}$ denotes the corresponding attention mask value. We then serialize extracted vectors into output/cleaned_cohort/clean_arabert_embeddings.npy (335×768) and output/cleaned_cohort/po1_strict_arabert_embeddings.npy (819×768).

5.3 Graph Feature Construction and Network Formulation

The node-feat matrix is assembled by the graph processing module (scripts/build_clean_dataset.py), which also ingests transmission interactions from Sanad Set.

5.3.1 Adjacency Matrix and Graph Serialization

The transmission network is constructed as an unweighted, undirected, NetworkX graph $G = (V, E)$. Each node index is mapped to an integer between 0 and $N-1 (N = 335)$. The 177 unique undirected teacher-student transmission pairs are stored as 354 directed message-passing edges (edge_index shape: 2×354) in PyTorch Geometric format and saved as output/cleaned_cohort/clean_graph.npz.

5.3.2 Extraction of Topological Measures

For each node $v_i \in V$, five structural properties are calculated. The number of total appearances (respectively, number of total appearances across the corpus) shows how many times the narrator appears, taken from the set of all verifiable transmission chains. In-degree shows how many unique teachers the narrator learned narrating from, while out-degree shows how many unique students the narrator has narrated to. Net flow shows relative directionality

balance of transmission (i.e., in degree / out degree). Mean chain position shows the mean normalised position of the narrator in the chain (which begins at 0, at source, and ends at 1, at collector).

5.3.3 Biographical Metadata Processing

Three biographical demographic features are used from the Hadith-Viz cards:

i. **Death Year (AH):** The year of death according to the Hijrī calendar, with missing values imputed using training-fold generation medians.
ii. **Generation index (*Ṭabaqa*):** The type of chronological generation and encoded as an ordinal scalar by dividing it by 12. Generation indices (*ṭabaqa*) are extracted through a two-tier resolution pipeline that prioritises verified numerical ranks (tb) from Hadith-Viz cards, using regular expression pattern matching on raw Taqrīb al-Tahdhīb biographical text to resolve unpopulated entries.
iii. **Classical statement count:** The total number of classical earlier critic appraisal statements available for the narrator.

All eight tabular features were z-score normalised ($z = \frac{x-\mu}{\sigma}$) and fit only on the training data before being presented to the neural network.

5.4 Neural Architecture Details & Training Hyperparameters

All neural network models used in the 13-arm ablation suite (P1-B_MLP, G1, G2, P2-A, P2-B_Concat, P2-B_Attn, P2-C, P2-D) are coded in PyTorch and PyTorch Geometric, in scripts/run_clean_experiments.py.

Table 5.2: Neural Network Hyperparameter Configuration Across Experimental Arms

Hyperparameter Setting	Configuration Value	Methodological Rationale
Optimiser	Adam	Standard first-order stochastic gradient optimisation for neural classifiers.

Base Learning Rate	5×10^{-3}	Learning rate used to allow fast and stable convergence of the tabular and GNN layers.
Weight Decay (L_2 Regularisation)	1×10^{-4}	Limit the size of weights to maintain the stability of the variance of the gradient.
Training Epochs	100	Sufficient maximum capacity to converge, both on the tabular and GNN layers.
Early Stopping Strategy	Inner-train fold	Early stopping was trained on inner-train fold to learn in not overfitting to the test.
Loss Function	Weighted Cross-Entropy	Loss weighted with the inverse class frequency to handle the imbalance in classes.
Activation Function	ReLU	Rectified Linear Unit activation between the hidden layers.
Dropout Probability	$p = 0.3$	Used in the neural arms with dropout layers.
Number of dimensions in the hidden layer	64 dimensions	Used by the implemented neural experiment arms.

5.4.1 Inverse-Frequency Loss Weighting

The weighted cross-entropy loss function, with inverse class frequency weights in order to down-weight the major class, so that the neural models attend evenly to the minority reliability class (*Below Ṣadūq*, 5.97% of the multimodal cohort) are constructed as:

$$\mathcal{L} = -\sum_{c=1}^{C} w_c \cdot y_c \log(\hat{p}_c),\ w_c = \frac{N}{C \cdot N_c}$$

Equation 5.3

where N is the total number of training set samples, C is the total number of target classes ($C = 3$), N_c is the number of training set samples with class c, y_c is 1 if class c is present, otherwise 0, and $\hat{p}_c$ is the softmax predicted probability for class c.

5.5 Verification, Unit Tests, and Diagnostic Tools for the Pipeline

Before production benchmarking began, the pipeline was checked with four diagnostic unit tests to ensure its quality:

i. Target Leakage String Scan: An automatic testing script was used to make sure that none of the biographical text inputs contained target strings or attributions related to Ibn Ḥajar, al-Dhahabī, al-Mizzī, or post-classical texts in the clean_texts.json file. The scan found no such strings in the checked text inputs.

ii. Identity Partition Integrity: During the Identity Partition Integrity test, an assertion checked for all folds $k \in \{1, \ldots, 25\}$ that $training\ identity\ IDs \cap testing\ identity\ IDs = \emptyset$, therefore len$(V_{\text{train}} \cap V_{\text{test}}) = 0$.

iii. Graph Adjacency Symmetry Check: The Graph Adjacency Symmetry Check validated that the transmission adjacency matrix A satisfied the condition $A = A^T$ and that the diagonal contained zero self-loops prior to the normalisation of the convolution.

iv. Downstream *Isnād* Index Alignment: Candidate-chain retrieval and the guarded evaluation tools were checked for protocol readiness. Blinded review and eligibility assessment were performed to establish narrator identity for every link, independent grade validity, complete-chain coverage, and PO3 performance.

CHAPTER 6

RESULTS

6.1 Pre-Flight Unit Testing and Pipeline Diagnostic Results

Prior to assessing predictive models, four automated diagnostic unit tests were run to validate the data and ensure the data does not leak into our target:

a) **Target Leakage String Scan:** Run text-matching audit on all text input in output/cleaned_cohort/clean_texts.json for references to post-classical compilers (Ibn Ḥajar, ad-Dhahabī, al-Mizzī, al-Khazrajī, al-Ṣafadī) or reference to target verdicts. All 819 text-cohort entries and 335 multimodal-cohort entries were free of text leaks.

b) **Identity Partition Disjointness Check:** An assertion audit validated that training and testing narrator identities for all 25 cross-validation folds exhibited a definitively empty intersection:Identities not contaminated between train/test sets.

$$\text{len}(V_{\text{train}} \cap V_{\text{test}}) = 0$$

c) **Graph Adjacency Symmetry Check:** Transmission adjacency matrix passed the following matrix audit: $A = A^T$, has no self-connections on the main diagonal before the convolution step, and is consistent with the known 177 undirected links in the network (PyTorch Geometric has 354 directed message edges).

d) **PO3 Evaluation Readiness:** Candidate retrieval and the guarded evaluation workflow have been prepared. This software readiness check does not validate the identity of every narrator in a chain or the independence of the corresponding hadith grade.

6.2 PO1 Large-Scale Biographical Text Benchmark Results (N = 819)

Project Objective 1 (PO1) evaluates canonical narrator reliability tiers from classical earlier-critic statements under zero leakage. For the purpose of this experiment, only the 819 historical scholars for which verified *Taqrīb* grades and earlier-critic commentary confined to

the second to fourth centuries AH are evaluated (Table 6.1). All models are evaluated over 25 paired cross-validation runs (5-fold Stratified Cross-Validation over 5 random restarts).

Table 6.1: PO1 Large-Scale Biographical Text Benchmark Results (N = 819)

Model ID	Model Architecture	Macro-F1 (Mean ± SD)	95% Confidence Interval	Accuracy (%)	Quadratic Weighted κ
B0	Majority Baseline	**0.2702 ± 0.0006**	[0.2699, 0.2704]	68.13%	0.0000
B1	Stratified Random	**0.3261 ± 0.0398**	[0.3097, 0.3426]	50.84%	-0.0195
P1-A	Lexical TF-IDF LR	**0.6852 ± 0.0514**	[0.6640, 0.7064]	**80.59%**	**0.6697**
P1-B	AraBERT Semantic LR	**0.5635 ± 0.0373**	[0.5481, 0.5789]	69.99%	0.5008

6.2.1 PO1 Findings Summary

i. Both machine learning models beat substantially the majority baseline (B0: 0.2702 Macro-F1) as well as the stratified chance (B1: 0.3261 Macro-F1) demonstrating that earlier-critic reports have high predictive power.

ii. The lexical TF-IDF classifier (P1-A) leads the pack in the final evaluations with a Macro-F1 of **0.6852 ± 0.0514**, an overall accuracy of **80.59%**, as well as high ordinal agreement ($\kappa = 0.6697$).

iii. The pretrained classifiers using AraBERT (P1-B) obtained **0.5635 ± 0.0373** Macro-F1 and **69.99%** accuracy. Dense pretrained embeddings did not perform as well as regularised lexical n-grams at $\overline{\Delta} = -0.1218 (t = -11.4153, p < 0.0001)$.

6.3 PO2 Controlled Thirteen-Arm Multimodal Benchmark Results (N = 335)

Project Objective 2 (PO2) tests whether transmission network features provide predictive advantage over text alone when used in conjunction with biographical text. Performance for

all 13 model arms, tested on the reduced multimodal cohort is shown in Table 6.2 ($N = 335$ nodes, 177 unique undirected relationships representing 354 directed message edges in PyG).

Table 6.2: Benchmark Results Across All 13 Experimental Arms (N = 335, 25 Paired Folds)

Arm ID	Model Identifier	Model Architecture	Macro-F1 (Mean ± SD)	95% Confidence Interval	Accuracy (%)	Quadratic κ	Consensus F1 ($N = 180$)
B0	Majority Baseline	Constant Majority	**0.2881 ± 0.0000**	[0.2881, 0.2881]	76.12%	0.0000	0.3593
B1	Stratified Random	Empirical Sampling	**0.3272 ± 0.0601**	[0.3024, 0.3520]	61.25%	-0.0133	0.3050
B2	Name + Meta Probe	Logistic Regression	**0.3822 ± 0.0439**	[0.3641, 0.4003]	50.69%	0.0318	0.3436
P1-A	Lexical TF-IDF LR	Logistic Regression	**0.5608 ± 0.0703**	[0.5318, 0.5898]	**80.96 %**	**0.4932**	**0.5284**
P1-B	AraBERT Semantic LR	Logistic Regression	**0.4933 ± 0.0628**	[0.4674, 0.5192]	72.84%	0.4347	0.4478

P1-B_MLP	AraBERT Semantic MLP	2-layer MLP (Hidden: 64)	**0.4454 ± 0.0657**	[0.4183, 0.4725]	68.60%	0.3295	0.4037
G1	Graph Features MLP	2-layer MLP (Hidden: 64)	**0.3556 ± 0.0564**	[0.3323, 0.3788]	47.64%	0.0899	0.3030
G2	Graph GCN	2-layer GCN (Hidden: 64)	**0.3438 ± 0.0716**	[0.3142, 0.3733]	44.30%	0.0343	0.2755
P2-A	Early Fusion MLP	2-layer MLP (Hidden: 64)	**0.4810 ± 0.0684**	[0.4528, 0.5093]	68.36%	0.3602	0.4708
P2-B_Concat	Multimodal GCN Concat	2-layer GCN (Hidden: 64)	**0.4311 ± 0.0750**	[0.4002, 0.4621]	64.96%	0.2157	0.3979
P2-B_Attn	Multimodal GCN Attention	Cross-Modal GNN Attention	**0.4478 ± 0.0670**	[0.4202, 0.4755]	66.27%	0.2549	0.4271
P2-C	Topology Isolation MLP	2-layer MLP (Hidden: 64)	**0.4531 ± 0.0727**	[0.4231, 0.4831]	67.22%	0.3318	0.4270

P2-D	Metadata Isolation MLP	2-layer MLP (Hidden: 64)	**0.4315 ± 0.0796**	[0.3987, 0.4644]	65.97%	0.2919	0.3980

Table 6.3: Per-Class F1 Breakdown Across Tiers on Condensed Cohort (N = 335)

Arm ID	**Model Identifier**	**Below Ṣadūq ($N = 20$)**	**Ṣadūq ($N = 60$)**	**Thiqah ($N = 255$)**
B0	Majority Baseline	0.0000	0.0000	0.8644
B1	Stratified Random	0.0294	0.1952	0.7570
B2	Name + Meta Probe	0.1785	0.3415	0.6267
P1-A	Lexical TF-IDF LR	0.1630	**0.6170**	**0.9024**
P1-B	AraBERT Semantic LR	0.1885	0.4352	0.8562
P1-B_MLP	AraBERT Semantic MLP	0.1548	0.3545	0.8269
G1	Graph Features MLP	0.1630	0.2865	0.6173
G2	Graph GCN	0.1559	0.3248	0.5506
P2-A	Early Fusion MLP	**0.2682**	0.3623	0.8126
P2-B_Concat	Multimodal GCN Concat	0.1609	0.3585	0.7739
P2-B_Attn	Multimodal GCN Attention	0.1975	0.3554	0.7906
P2-C	Topology Isolation MLP	0.2178	0.3321	0.8093

P2-D	Metadata Isolation MLP	0.1779	0.3170	0.7998

6.3.1 PO2 Findings Summary:

i. Model P2-A (Early Fusion MLP) obtained the highest multimodal Macro-F1 (**0.4810 ± 0.0684**), and obtained the highest discrimination on the minority class (*Below Ṣadūq* F1 = **0.2682**).

ii. Relational GCN message passing (P2-B_Concat: 0.4311 Macro-F1) hurt performance compared to non-relational tabular fusion (P2-A: 0.4810 Macro-F1). Transmitting representations along mismatched teacher-student edges resulted in feature over-smoothing.

iii. Non-relational topological metrics alone (G1: 0.3556 Macro-F1) outperformed pure graph convolution (G2: 0.3438 Macro-F1), confirming that network position adds some structural information but cannot stand in for text evaluation.

6.4 Formal Statistical Hypothesis Testing Decisions

Table 6.4 describes the formal pairwise hypothesis testing decisions on all 25 paired cross-validation folds ($df = 24$, $\alpha = 0.05$).

Table 6.4: Paired Two-Tailed Hypothesis Testing Results Across 25 Paired Folds (df = 24)

Research Hypothesis	**Paired Comparison (Arm A vs. Arm B)**	**Mean Difference $\bar{\Delta}$**	**95% Confidence Interval**	**Paired t-statistic**	**p-value**	**Empirical Decision**
$H_{0,1}$**: Semantic vs Lexical (N=819)**	P1-B (AraBERT) vs P1-A (TF-IDF)	**-0.1218**	[-0.1438, -0.0998]	$t = -11.4153$	$p < 0.0001$	**Reject H_0(TF-IDF leads)**
$H_{0,1}$**: Semantic**	P1-B (AraBERT) vs	**-0.0675**	[-0.1051, -0.0299]	$t = -3.7035$	$p = 0.00111$	**Reject H_0(TF-**

vs Lexical (N=335)	P1-A (TF-IDF)					**IDF leads)**
$H_{0,2a}$**: Message Passing vs Feature MLP**	G2 (Graph GCN) vs G1 (Features MLP)	**-0.0118**	[-0.0436, +0.0200]	$t = -0.7667$	$p = 0.45076$	**Fail to Reject** H_0**(No edge gain)**
$H_{0,2b-c}$**: GCN Concat vs Early Fusion**	P2-B_Concat vs P2-A (Early Fusion)	**-0.0499**	[-0.0918, -0.0080]	$t = -2.4595$	$p = 0.02150$	**Reject** H_0**(P2-A higher)**
$H_{0,2b-a}$**: Attention GNN vs Early Fusion**	P2-B_Attn vs P2-A (Early Fusion)	**-0.0332**	[-0.0656, -0.0008]	$t = -2.1130$	$p = 0.04519$	**Reject** H_0**(P2-A higher)**

6.4.1 Hypothesis Outcomes Summary

a) Resolution of RQ1 ($H_{0,1}$): Pretrained AraBERT is not better than standardised lexical TF-IDF. Across both cohorts ($N = 819$, $t = -11.4153$, $p < 0.0001$; $N = 335$, $p = 0.00111$), lexical TF-IDF demonstrates statistically significant superiority.

b) Resolution of RQ2: The G2 versus G1 comparison did not show a statistically significant difference ($p = 0.45076$). Both multimodal GNN comparisons differed significantly from P2-A on the paired folds (P2-B_Concat: $p = 0.02150$; P2-B_Attn: $p = 0.04519$), with P2-A having the higher mean Macro-F1 in each comparison.

6.5 PO3 Downstream Chain Validation Results

A preliminary PO3 exercise evaluated the candidate retrieval and weakest-link scoring workflow. The retrieval package contained 800 queries, and the known target was retrieved within the top 50 candidates for all 691 originally accepted queries. This checks candidate retrieval coverage; it does not measure grading performance.

In a separate exploratory scoring run, 1,200 source queries produced 657 aligned rows before deduplication. After conflicting matches, parse failures, and records with insufficient narrator coverage were excluded, 181 chain records remained. Each record's score used majority-voted out-of-fold predictions from the inductive GCN and weakest-link aggregation over its matched cohort narrators. The descriptive Spearman correlation between predicted scores and mapped reference grades was **$\rho = 0.054$**.

This is an exploratory result, not an evaluation under the finalised PO3 protocol: matching was automated, records could have unresolved narrators, and grades were not independently reviewed. In the subsequent review-gated preflight, none of the 800 queries had completed review, so no records passed the eligibility checks and no protocol-conformant PO3 association could be calculated. The preliminary correlation therefore does not establish that the finalized method agrees with traditional gradings.

CHAPTER 7

CONCLUSION

7.1 Fulfillment of Research Objectives

This study formulated and evaluated a zero-leakage computational framework for quantifying narrator reliability in *ʿIlm ar-Rijāl*. All three project objectives established in Chapter 1 were systematically achieved:

i. Fulfillment of PO1 (Large-Scale Text Representation Benchmark): A dual temporal firewall purged all 9th-century compilation verdicts and later syntheses from biographical entries, isolating commentary strictly from critics of the second to fourth centuries AH. Across 819 canonical scholars evaluated over 25 paired cross-validation folds, regularised lexical TF-IDF logistic regression achieved a Macro-F1 of **0.6852 ± 0.0514**, an overall accuracy of **80.59%**, and a Quadratic Weighted Kappa of $\kappa = 0.6697$. This performance significantly exceeded the majority baseline (Macro-F1: 0.2702) and dense AraBERT semantic representations (Macro-F1: **0.5635 ± 0.0373**, $p = 3.49 \times 10^{-11}$).

ii. Fulfillment of PO2 (Controlled Multimodal Graph and Text Fusion): A 13-arm controlled matrix was benchmarked on the condensed multimodal cohort ($N = 335$ scholars, 177 unique transmission ties representing 354 PyTorch Geometric directed message edges). Tabular Early Fusion MLP (Arm P2-A) achieved the top multimodal Macro-F1 score of **0.4810 ± 0.0684** and delivered the strongest discrimination on the minority class (*Below Ṣadūq* F1 = **0.2682**). Graph convolutional message passing across transmission edges yielded no performance advantage (GCN Macro-F1: **0.4311 ± 0.0750**, $p = 0.0215$ relative to early fusion).

iii. Progress of PO3 (Downstream Isnād Chain Validation): As of this writing, only a preliminary implementation of the new PO3 method has run. The preliminary

correlation therefore does not establish that the finalised method agrees with traditional gradings yet.

7.2 Summary of Scientific and Theoretical Contributions

The empirical results yield three decisive theoretical contributions to computational Hadith studies and applied machine learning:

7.2.1 Lexical Specificity vs. Contextual Embeddings in Classical Arabic

Classical *rijāl* criticism employs a formulaic vocabulary developed by early hadith experts and critics (*thiqah*, *ṣadūq*, *ḍa ʿīf*, *laysa bihi ba ʾs*, *munkar al-ḥadīth*). Regularised linear classifiers operating on character and word n-grams capture these discriminative lexical markers directly. Pretrained contextual transformers like AraBERT, trained predominantly on Modern Standard Arabic news corpora, disperse attention weights over ceremonial honorifics (*tarāḥum*), diluting the concentrated evaluative signals of technical terminology.

7.2.2 The Transmission Heterophily Discovery

A primary finding of this investigation is the failure of Graph Neural Network message passing on Hadith transmission networks. Standard Graph Convolutional Networks execute normalised neighbourhood feature averaging:

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{\tilde{d}_i \tilde{d}_j}} h_j^{(l)} W^{(l)} \right)$$

This mechanism assumes homophily, where linked nodes share identical classes ($y_i \approx y_j$). In Hadith transmission, the network displays pronounced heterophily ($h = 0.4107$). Eminent masters such as Sufyān ath-Thawrī or Imām Mālik delivered public lectures attended by hundreds of students with varied competence, ranging from rigorous scholars to weak transmitters. Averaging node representations across discordant teacher-student edges causes feature over-smoothing, dragging weak student representations toward the master's *Thiqah* centroid and degrading classification performance.

7.3 Scholarly Ambiguity Analysis: Ibn Ḥajar vs. ad-Dhahabī Cases

The computational models were tested against historical cases where ninth-century synthesizer Ibn Ḥajar al-ʿAsqalānī (*Taqrīb al-Tahdhīb*) and eighth-century synthesizer ad-

Dhahabī (*al-Kāshif*) diverged in their canonical reliability tiers. Within the multimodal cohort ($N = 335$), 238 scholars possessed evaluations from both authorities:

a) Consensus Set: 180 scholars (75.6%) received identical reliability tiers from both authorities.

b) Disagreement Set: 58 scholars (24.4%) exhibited divergent classifications:

- In 38 cases, Ibn Ḥajar assigned *Ṣadūq* (Tier 1) while ad-Dhahabī assigned *Thiqah* (27 cases) or *Below Ṣadūq* (11 cases).
- In 18 cases, Ibn Ḥajar assigned *Thiqah* (Tier 2) while ad-Dhahabī assigned *Ṣadūq* (11 cases) or *Below Ṣadūq* (7 cases).
- In 2 cases, Ibn Ḥajar assigned *Below Ṣadūq* (Tier 0) while ad-Dhahabī assigned *Ṣadūq*.

Table 7.1: Illustrative Divergence Cases and Model Probability Distributions

ID	Narrator Name	Ibn Ḥajar Grade	Adh-Dhahabī Grade	Early Quotes	Model Prediction	Diagnostic Interpretation
1	Abū Bakr ibn Abī Shaybah	*Thiqah Ḥāfiẓ* (Tier 2)	*Ṣadūq* (Tier 1)	42	*Thiqah* ($p = 0.81$)	Ibn Ḥajar highlights his monumental work *Muṣannaf*; Adh-Dhahabī notes rare isolated slips. Model reflects overwhelming accreditation volume.
14	ʿAbd al-Ḥamīd ibn Jaʿfar	*Ṣadūq yakhṭi'* (Tier 1)	*Thiqah* (Tier 2)	53	*Ṣadūq* ($p = 0.58$)	Ibn Maʿīn endorsed him while Abū Ḥātim ar-Rāzī noted memory slips. Model probability split ($p_{\text{Thiqah}} = 0.39, p_{\text{Ṣadūq}} = 0.58$) mirrors historical divide.
30	Hishām ibn ʿAmmār	*Ṣadūq taghayyara* (Tier 1)	*Thiqah* (Tier 2)	65	*Ṣadūq* ($p = 0.62$)	Eminent preacher whose memory deteriorated in extreme old age. Model captures early qualifications regarding late transmissions.
39	Ḥammād ibn Salamah	*Thiqah ʿābid* (Tier 2)	*Ṣadūq* (Tier 1)	204	*Thiqah* ($p = 0.74$)	Basran pillar; critics debated relative manuscript

						alterations. High volume of early endorsements anchors prediction to *Thiqah*.

When early critics were unanimous, model prediction confidence was high ($p > 0.80$). When early critics were divided, the model produced split posterior probabilities, demonstrating that the system serves as an objective diagnostic tool that reflects historical scholarly nuance rather than forcing artificial consensus.

REFERENCES

Ahmad, N., Ahmed, S., & Abdullah, M. (2025). Hadith grading & machine learning: Feasibility of automatic isnād-analysis. *AL-JAMEI Research Journal*, 3(2), 202–215. https://aljamei.com/index.php/ajrj/article/view/155

Ahmed, M., & Anas, M. (2025). Hadith narrators' biographies and their role in the reliability of transmission. *Contemporary Journal of Social Science Review*, 3(3), 139–155. https://doi.org/10.63878/cjssr.v3i3.1131

Al-Aʿẓamī, M. M. (1977). *Studies in Hadith methodology and literature*. American Trust Publications.

Al-ʿAsqalānī, I. Ḥ. (1984). *Tahdhīb al-Tahdhīb* (12 vols.). Dāʾirat al-Maʿārif al-Niẓāmiyyah.

Al-ʿAsqalānī, I. Ḥ. (1995). *Taqrīb al-Tahdhīb* (A. ʿAwwāmah, Ed.). Dār al-Rashīd.

Al-Dhahabī, S. al-D. (1990). *Dhikr man yuʿtamad qawluhu fī al-jarḥ wat-taʿdīl* (ʿA. Abū Ghuddah, Ed.). In *Arbaʿ rasāʾil fī ʿulūm al-ḥadīth* (pp. 158–213). Maktab al-Maṭbūʿāt al-Islāmiyyah.

Al-Dhahabī, S. al-D. (1992). *Al-Kāshif fī Maʿrifat Man Lahu Riwāyah fī al-Kutub as-Sittah* (M. ʿAwwāmah, Ed., 2 vols.). Dār al-Qiblah li-al-Thaqāfah al-Islāmiyyah.

Al-Judaiʿ, ʿA. (2003). *Taḥrīr ʿUlūm al-Ḥadīth* (2 vols.). Muʾassasat al-Rayyān.

Al-Luknawī, ʿA. al-Ḥ. (2004). *Al-Rafʿ wat-takmīl fī al-jarḥ wat-taʿdīl* (ʿA. Abū Ghuddah, Ed., 8th ed.). Maktab al-Maṭbūʿāt al-Islāmiyyah.

Al-Mizzī, Y. (1980). *Tahdhīb al-Kamāl fī Asmāʾ ar-Rijāl* (B. ʿA. Maʿrūf, Ed., 35 vols.). Muʾassasat al-Risālah.

Antoun, W., Baly, F., & Hajj, H. (2020). AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT4)* (pp. 9–15). European Language Resources Association. https://aclanthology.org/2020.osact-1.2/

A'thoina, I., Hasballah, F., & Fadillah, N. A. (2025). Bridging revelation and algorithm: Integrating artificial intelligence and malakah in the epistemology of ḥadīth criticism. *Digital Muslim Review*, 3(1), 40–57. https://digitalmuslimreview.or.id/index.php/dmr/article/view/38

Brown, J. A. C. (2009). *Hadith: Muhammad's legacy in the medieval and modern world*. Oneworld Publications.

Ghriss, A. (2025). Crowdsourced information authentication: A graph-based model from the science of Hadith. In *Proceedings of the 4th Muslims in ML Workshop at ICML 2025* (PMLR Vol. 267). OpenReview.net. https://openreview.net/forum?id=F7CgAEPvAv

Inoue, G., Alhafni, B., Baimukan, N., Bouamor, H., & Habash, N. (2021). The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop (WANLP 2021)* (pp. 92–104). Association for Computational Linguistics. https://aclanthology.org/2021.wanlp-1.10/

Ismail, N. (2024). The role of Jarh wa Ta'dil in hadith authentication. *El-Sunan: Journal of Hadith and Religious Studies*, 2(1), 70–79. https://doi.org/10.22373/el-sunan.v2i1.5510

Kamali, M. H. (2014). *A textbook of Hadith studies: Authenticity, compilation, classification and criticism of Hadith*. Kube Publishing Ltd.

Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. In *Proceedings of the 5th International Conference on Learning Representations (ICLR 2017)*. https://openreview.net/forum?id=SJU4ayYgl

Lim, D., Hohne, F., Li, X., Huang, S. L., Gupta, V., Bhalerao, O., & Lim, S. N. (2021). Large scale learning on non-homophilous graphs: New benchmarks and strong simple methods. *Advances in Neural Information Processing Systems*, 34, 20887–20902. https://papers.neurips.cc/paper_files/paper/2021/hash/ae816a80e4c1c56caa2eb4e1819cbb2f-Abstract.html

Mahmoud, S., Saif, O., Nabil, E., Abdeen, M., ElNainay, M., & Torki, M. (2022). AR-Sanad 280K: A novel 280K artificial sanads dataset for Hadith narrator disambiguation. *Information*, 13(2), 55. https://doi.org/10.3390/info13020055

Mahmoud, S., Nabil, E., Saif, O., & Torki, M. (2024). Narrator identification by querying Sanad graph and utilizing the NarratorsKG on AR-Sanad 280K-v2 dataset. *Neural Computing and Applications*, 36(36), 23169–23180. https://doi.org/10.1007/s00521-024-10194-2

McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27(1), 415–444. https://doi.org/10.1146/annurev.soc.27.1.415

Mghari, M., Bouras, O., & El Hibaoui, A. (2022). Sanadset 650K: Data on Hadith narrators. *Data in Brief*, 44, 108540. https://doi.org/10.1016/j.dib.2022.108540

Mghari, M., Bouras, O., & El Hibaoui, A. (2024). Narrator2Vec: An efficient narrator representation in Hadith literature using word embedding. *Arabian Journal for Science and Engineering*, 49(3), 4479–4494. https://doi.org/10.1007/s13369-023-08224-7

Mosa, M. A. (2025). Synergizing structure and semantics: A knowledge graph-transformer framework for narrator disambiguation in hadith networks. *Digital Scholarship in the Humanities*, 40(4), 1085–1100. https://doi.org/10.1093/llc/fqaf088

Muther, R., Smith, D. A., & Savant, S. (2023). From networks to named entities and back again: Exploring classical Arabic isnād networks. *Journal of Historical Network Research*, 8(1), 1–20. https://doi.org/10.25517/jhnr.v8i1.135

Najeeb, M. M. A. (2022). A Hidden Markov Model-based tagging approach for Arabic isnads of Hadiths. *Mathematical Problems in Engineering*, 2022, 1–12. https://doi.org/10.1155/2022/7160509

Oumer, J. M. (2025). *Relation extraction for graph-based representation of Hadith isnad* [Master's thesis, Università ta' Malta]. https://www.um.edu.mt/library/oar/handle/123456789/137629

Saloot, M. A., Idris, N., Mahmud, R., Ja'afar, S., Thorleuchter, D., & Gani, A. (2016). Hadith data mining and classification: A comparative analysis. *Artificial Intelligence Review*, 46(1), 113–128. https://doi.org/10.1007/s10462-016-9458-x

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information*

Processing Systems (NeurIPS 2017), 30, 5998–6008. https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph attention networks. In *Proceedings of the 6th International Conference on Learning Representations (ICLR 2018)*. https://openreview.net/forum?id=rJXMpikCZ

Zhu, J., Yan, Y., Zhao, L., Heimann, M., Akoglu, L., & Koutra, D. (2020). Beyond homophily in graph neural networks: Current limitations and effective designs. *Advances in Neural Information Processing Systems*, 33, 7793–7804. https://proceedings.neurips.cc/paper/2020/hash/58ae23d878a47004366189884c2f8440-Abstract.html

APPENDICES